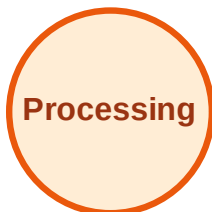


BFS Traversal

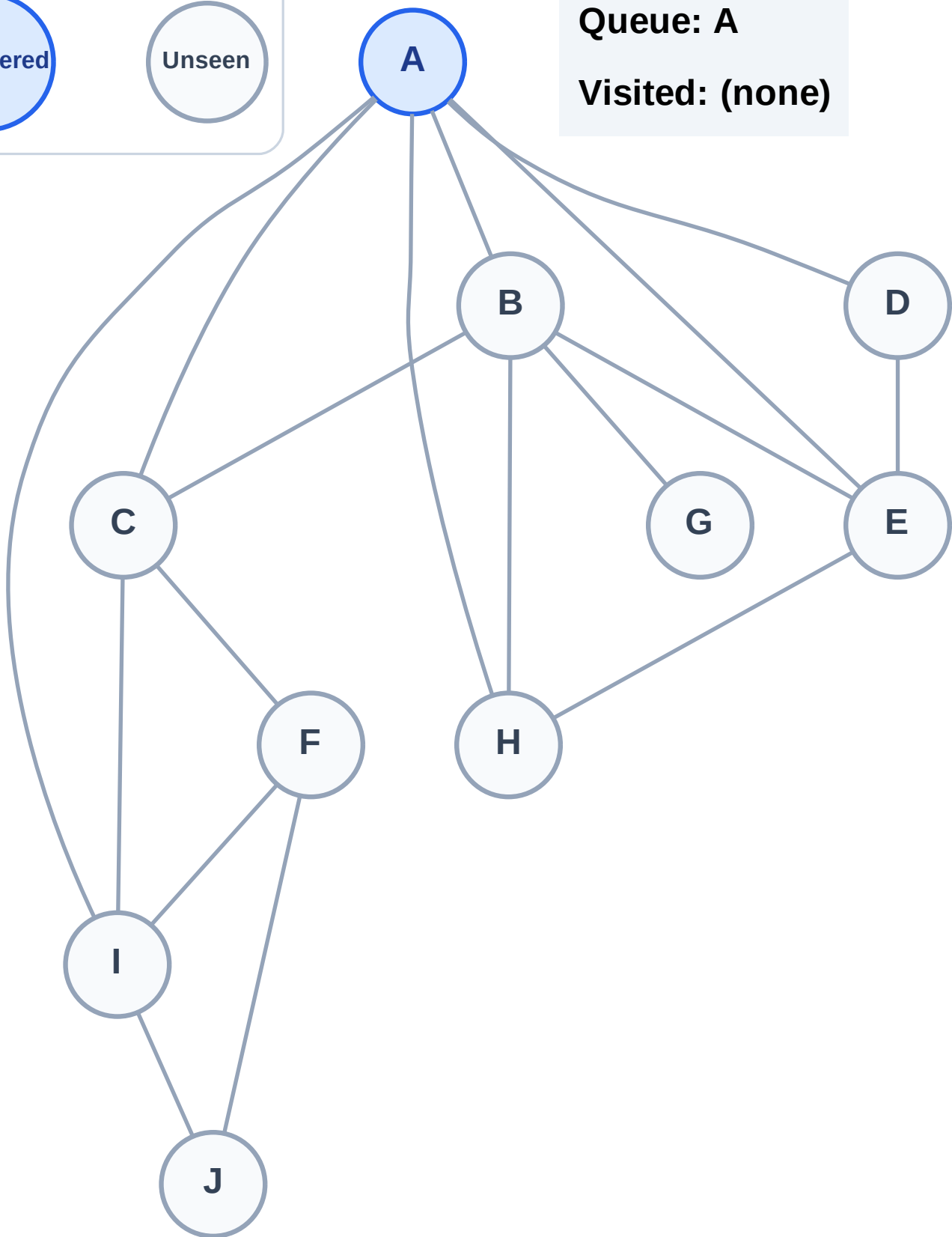
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

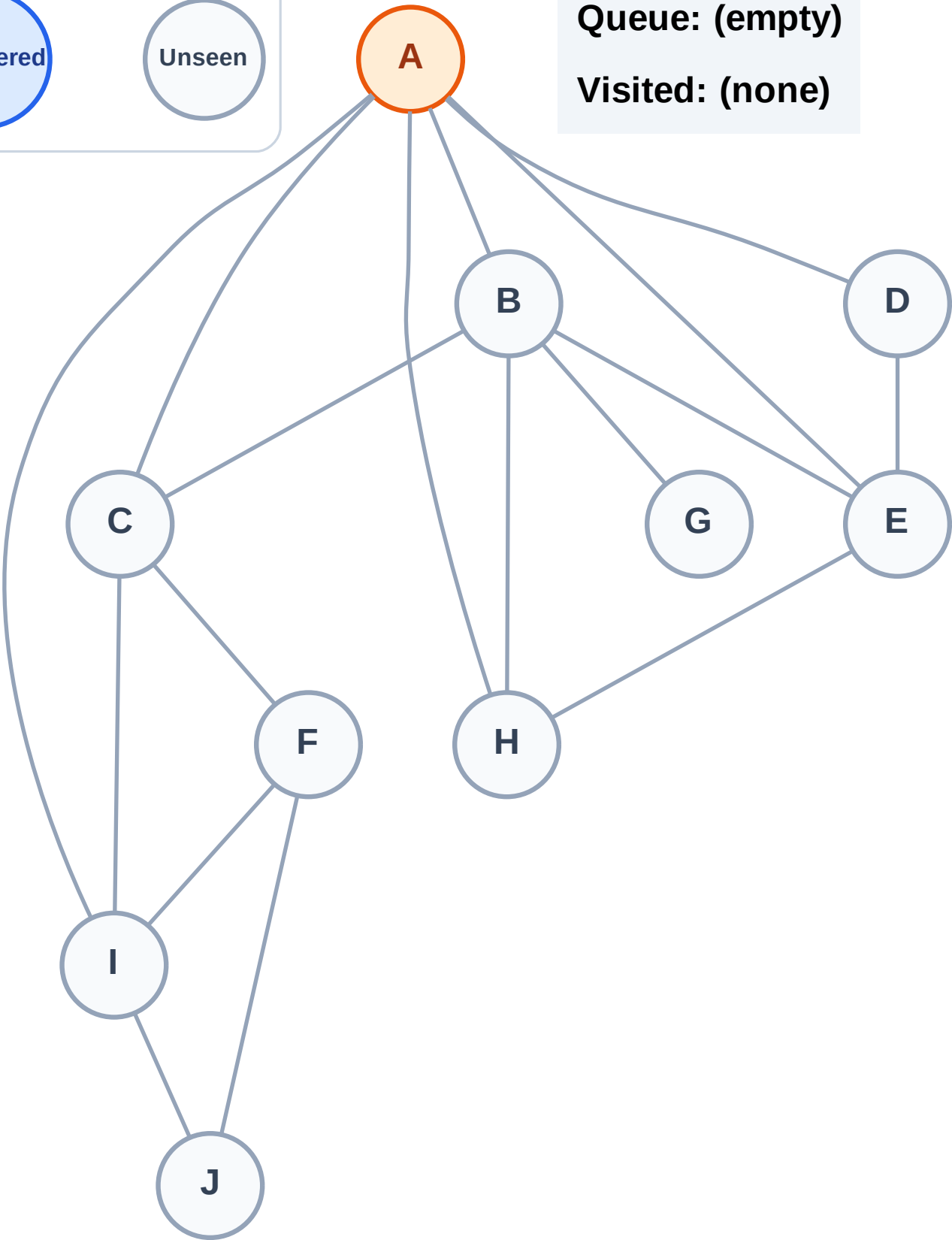
Complete

Processing

Discovered

Unseen

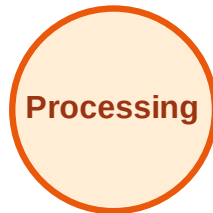
Queue: (empty)
Visited: (none)



BFS Traversal

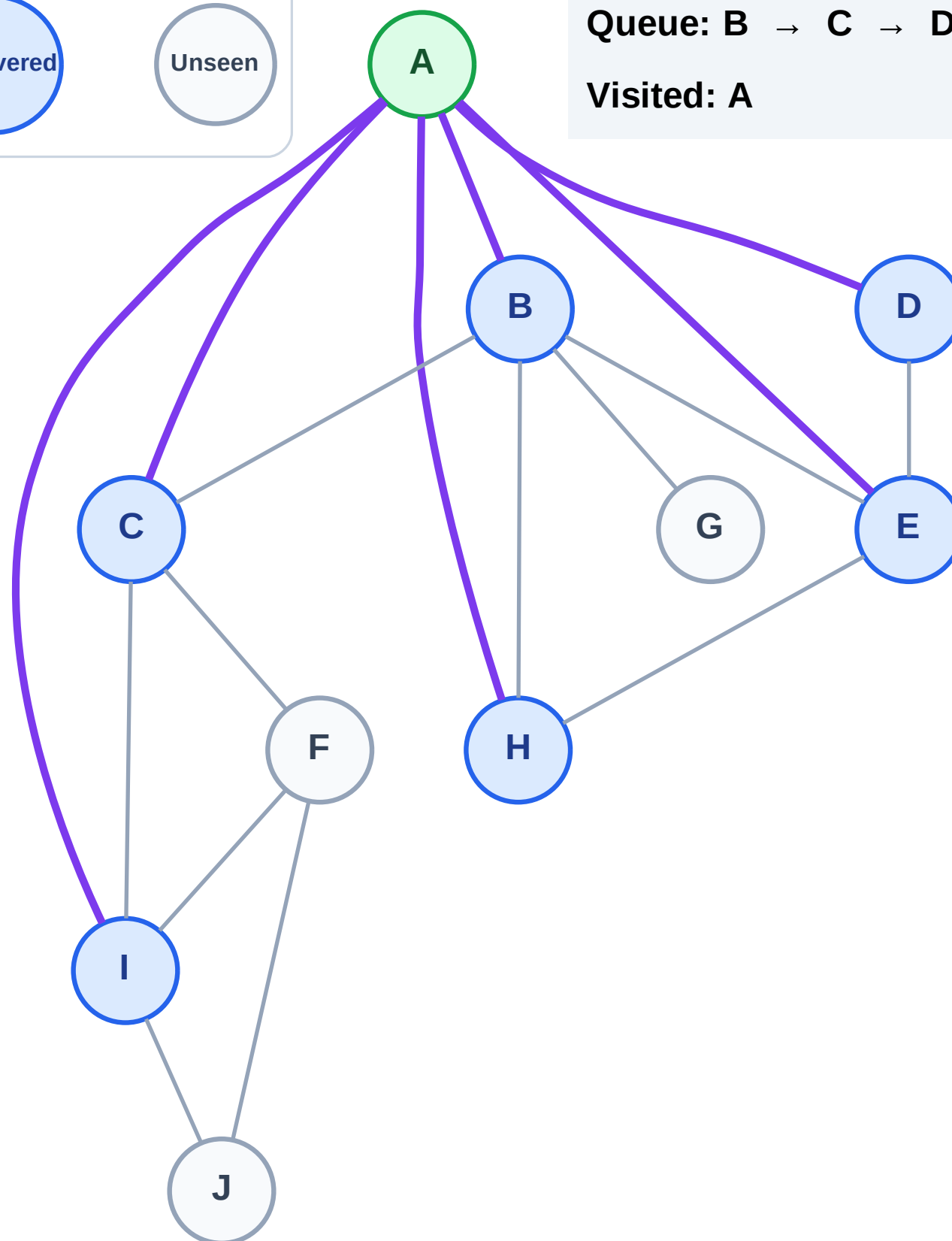
Step 3: Discover B, C, D, E, H, I from A

Legend



Queue: B → C → D → E → H → I

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

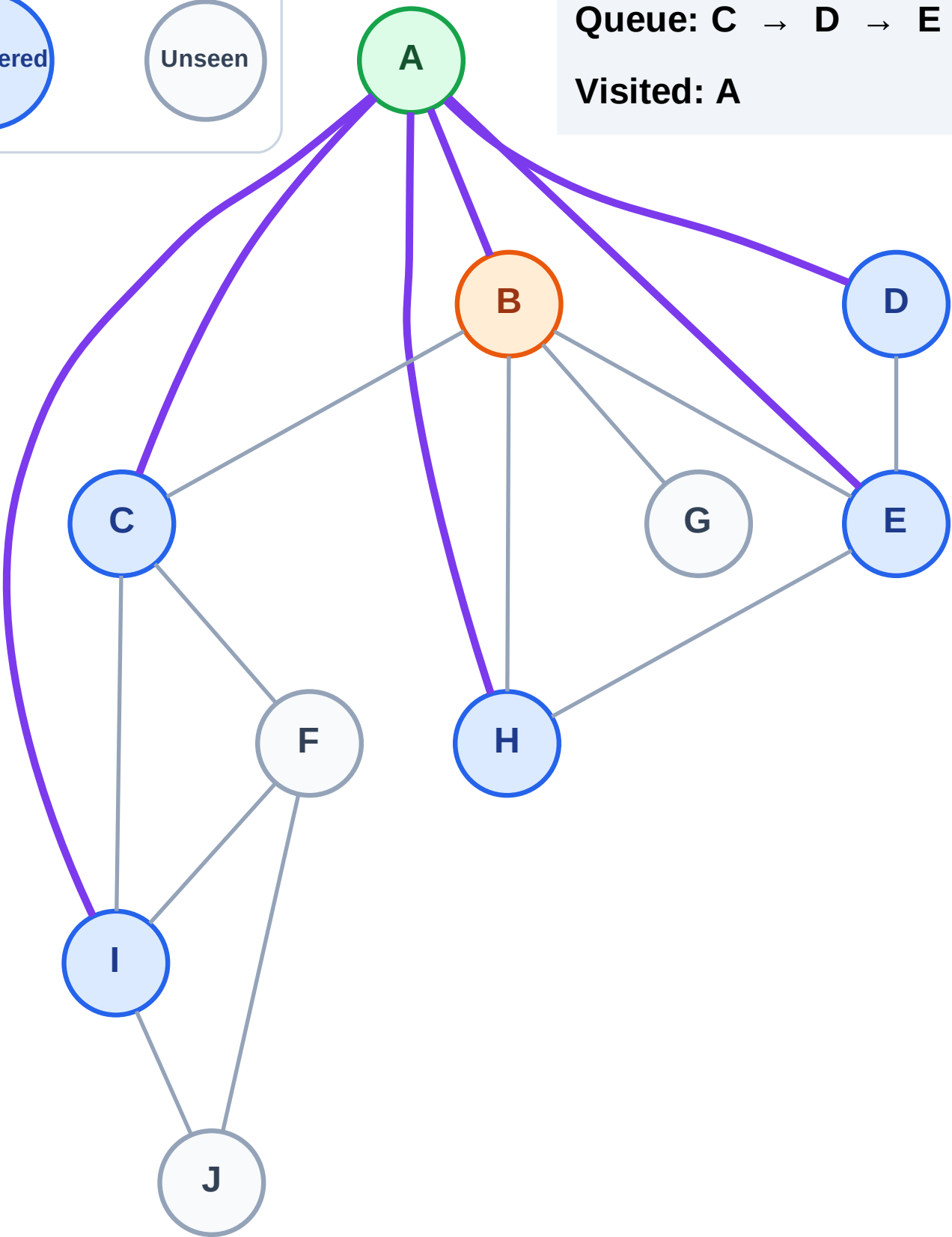
Processing

Discovered

Unseen

Queue: C → D → E → H → I

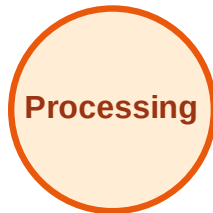
Visited: A



BFS Traversal

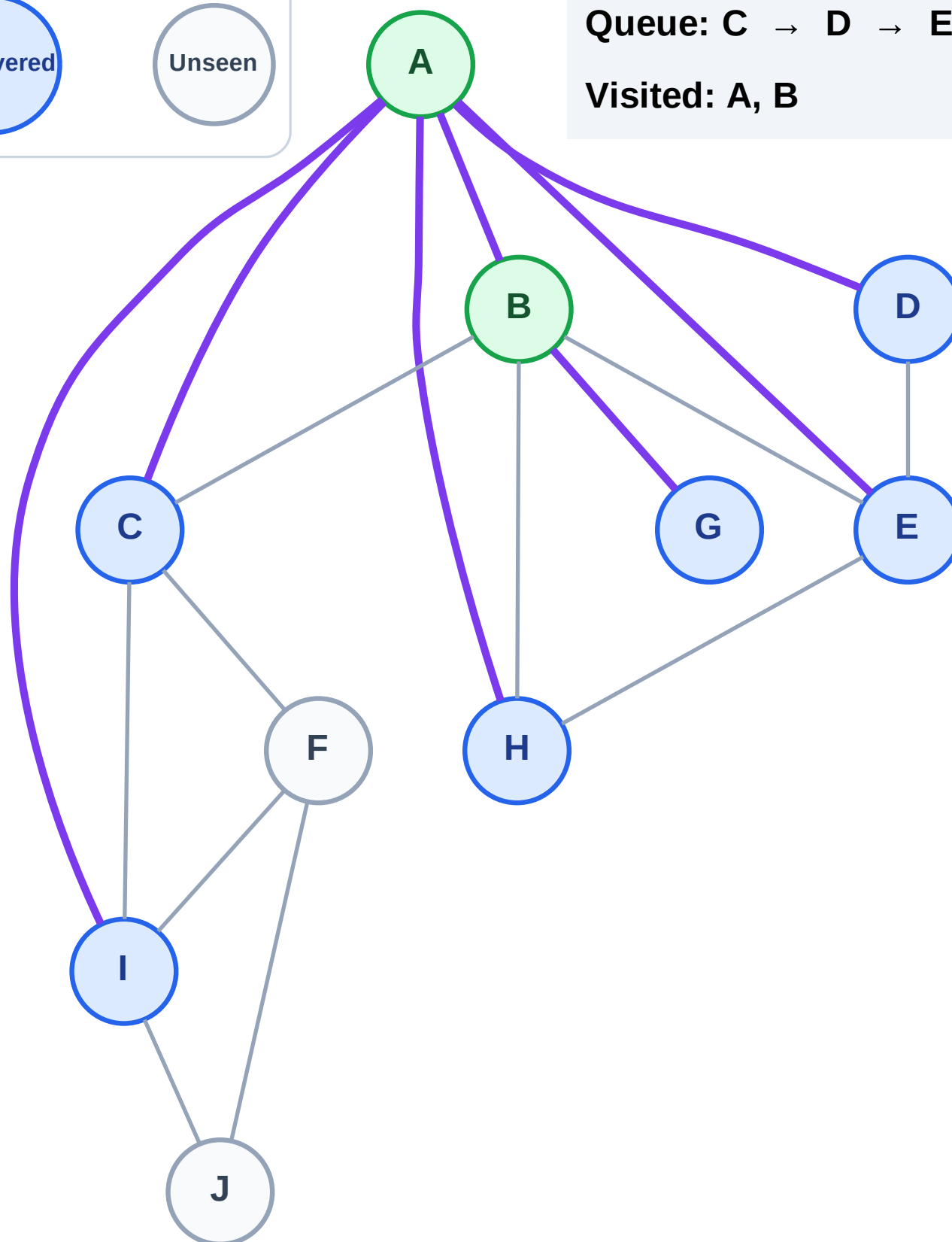
Step 5: Discover G from B

Legend



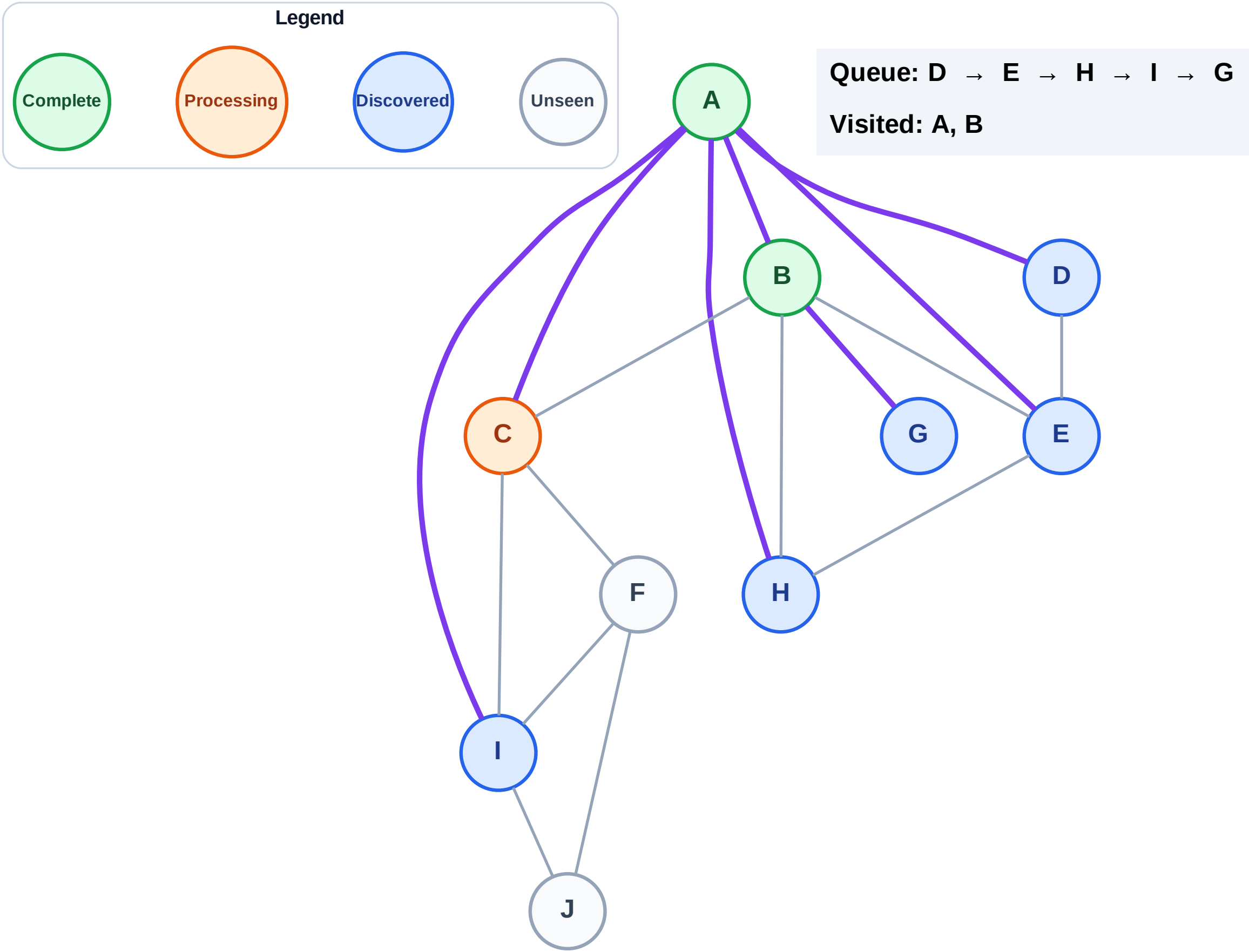
Queue: C → D → E → H → I → G

Visited: A, B



BFS Traversal

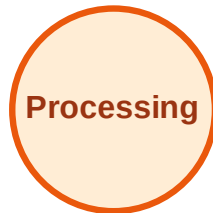
Step 6: Process node C



BFS Traversal

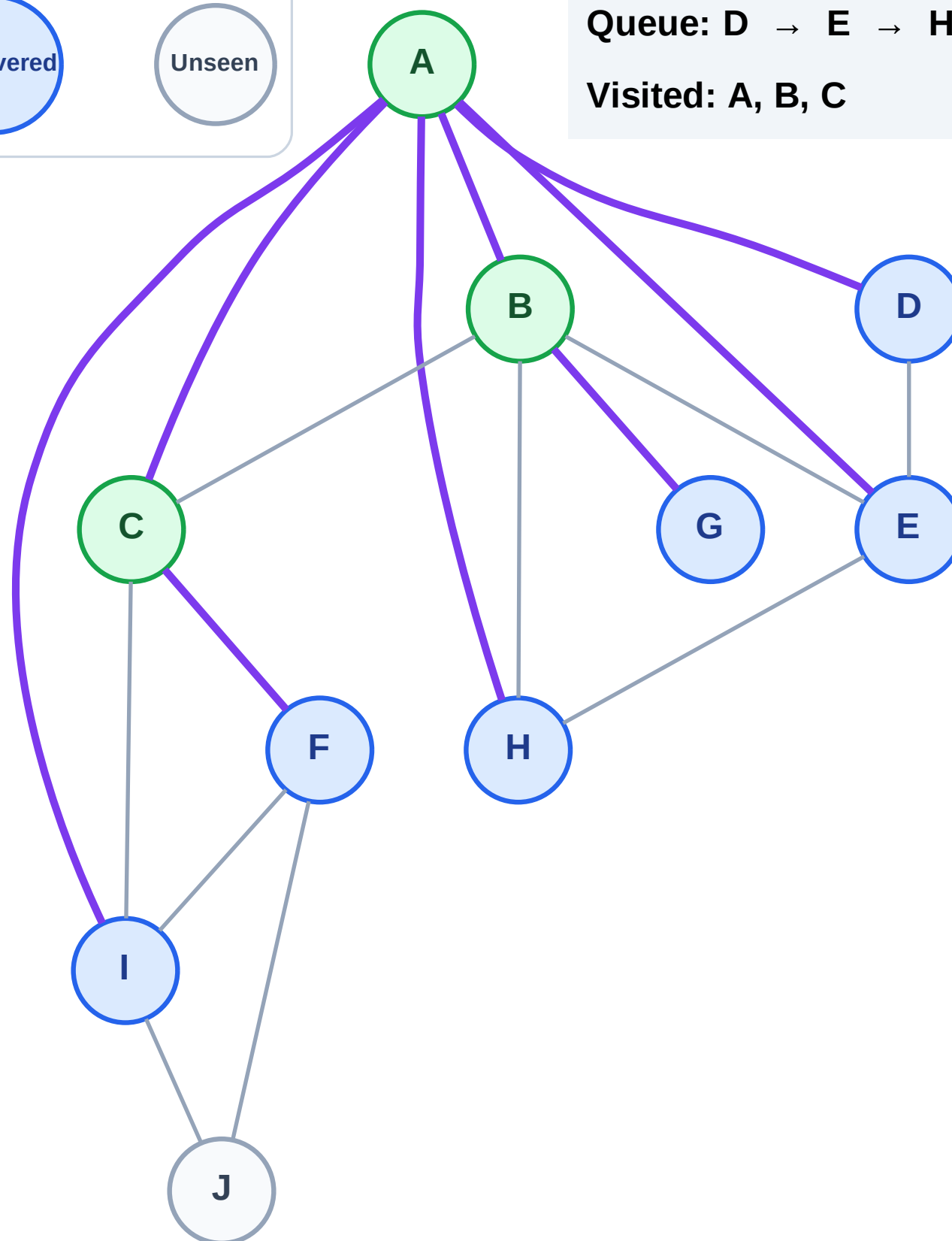
Step 7: Discover F from C

Legend



Queue: D → E → H → I → G → F

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

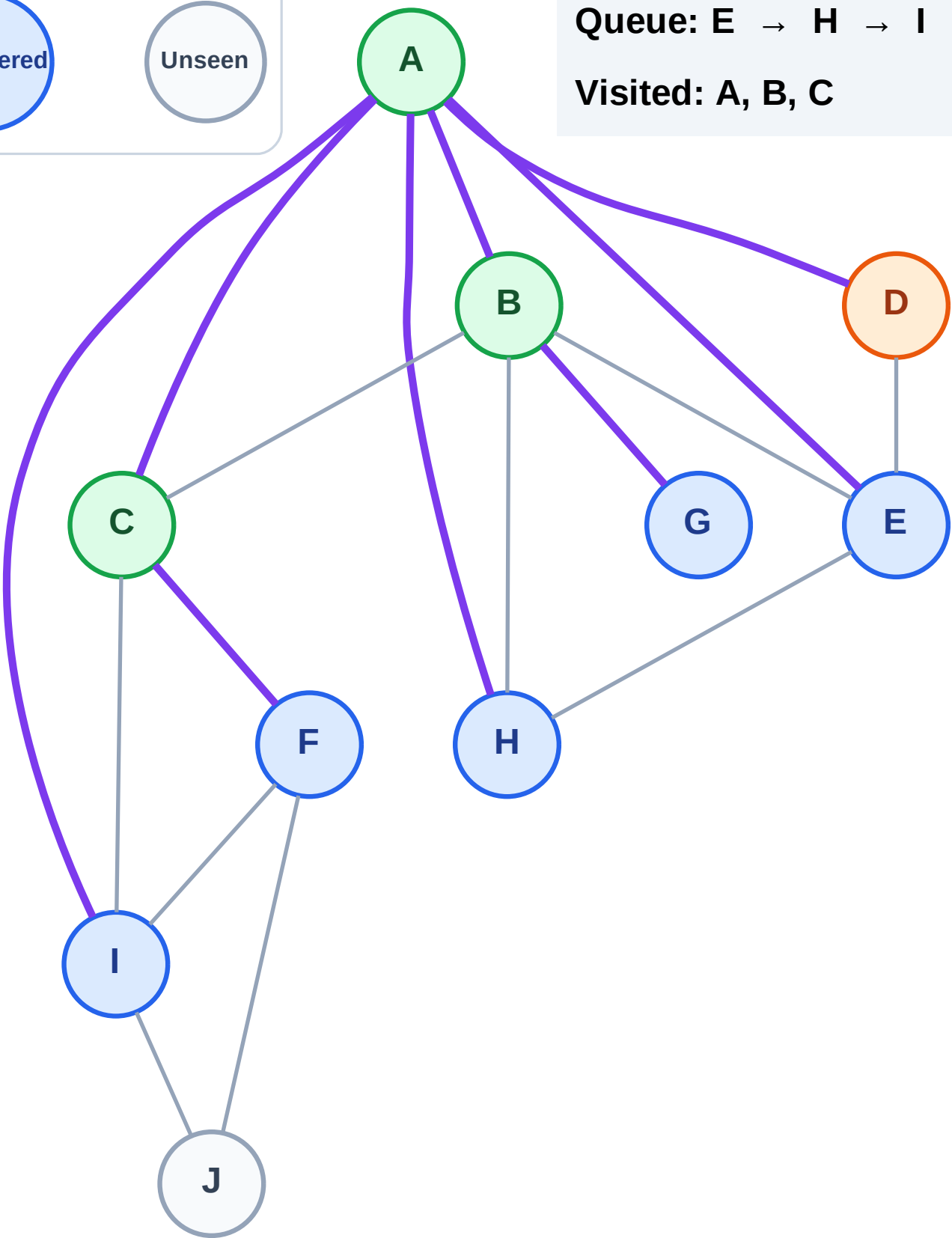
Processing

Discovered

Unseen

Queue: E → H → I → G → F

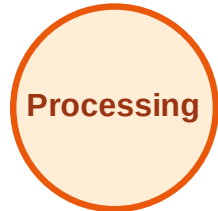
Visited: A, B, C



BFS Traversal

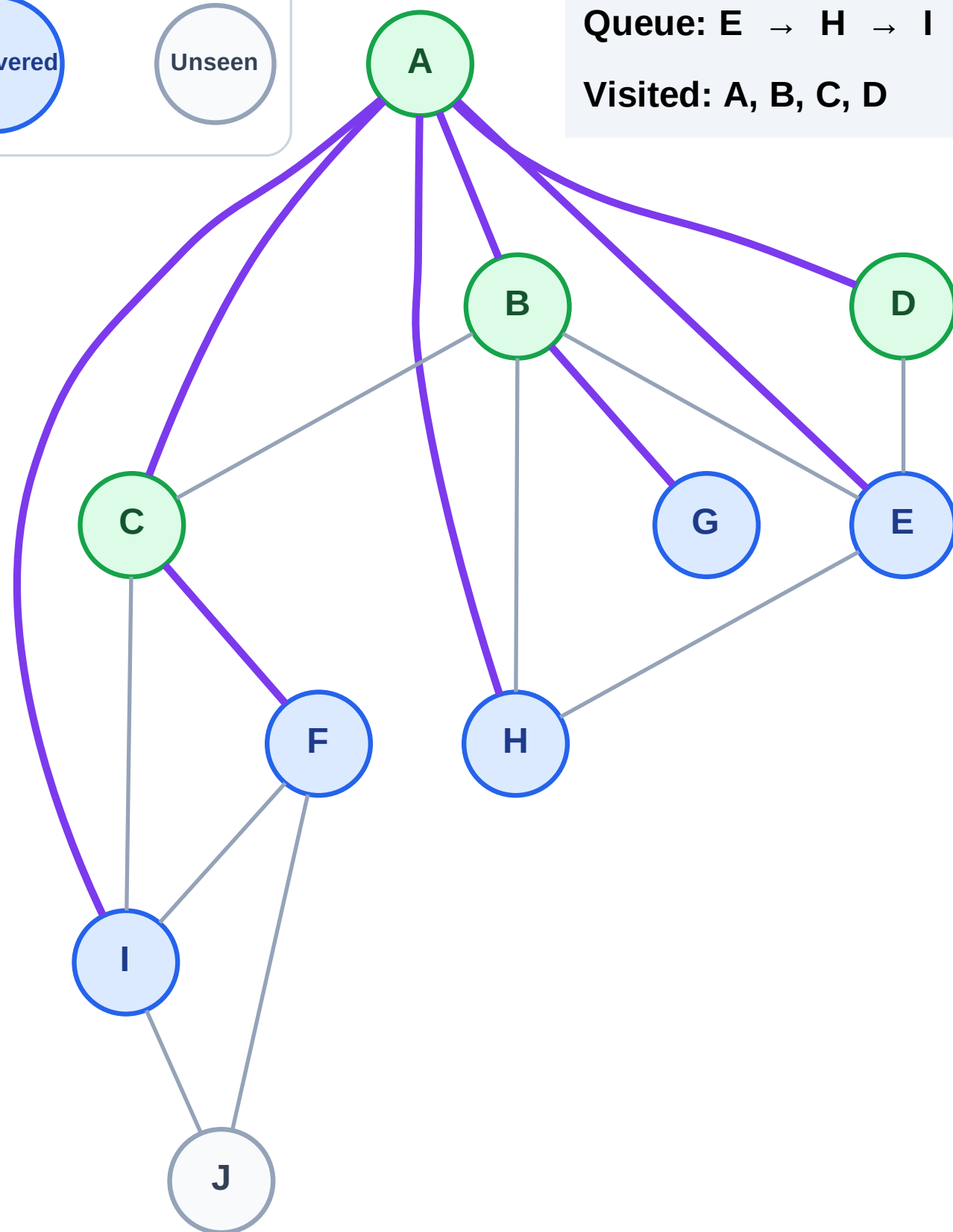
Step 9: No new neighbors from D

Legend



Queue: E → H → I → G → F

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

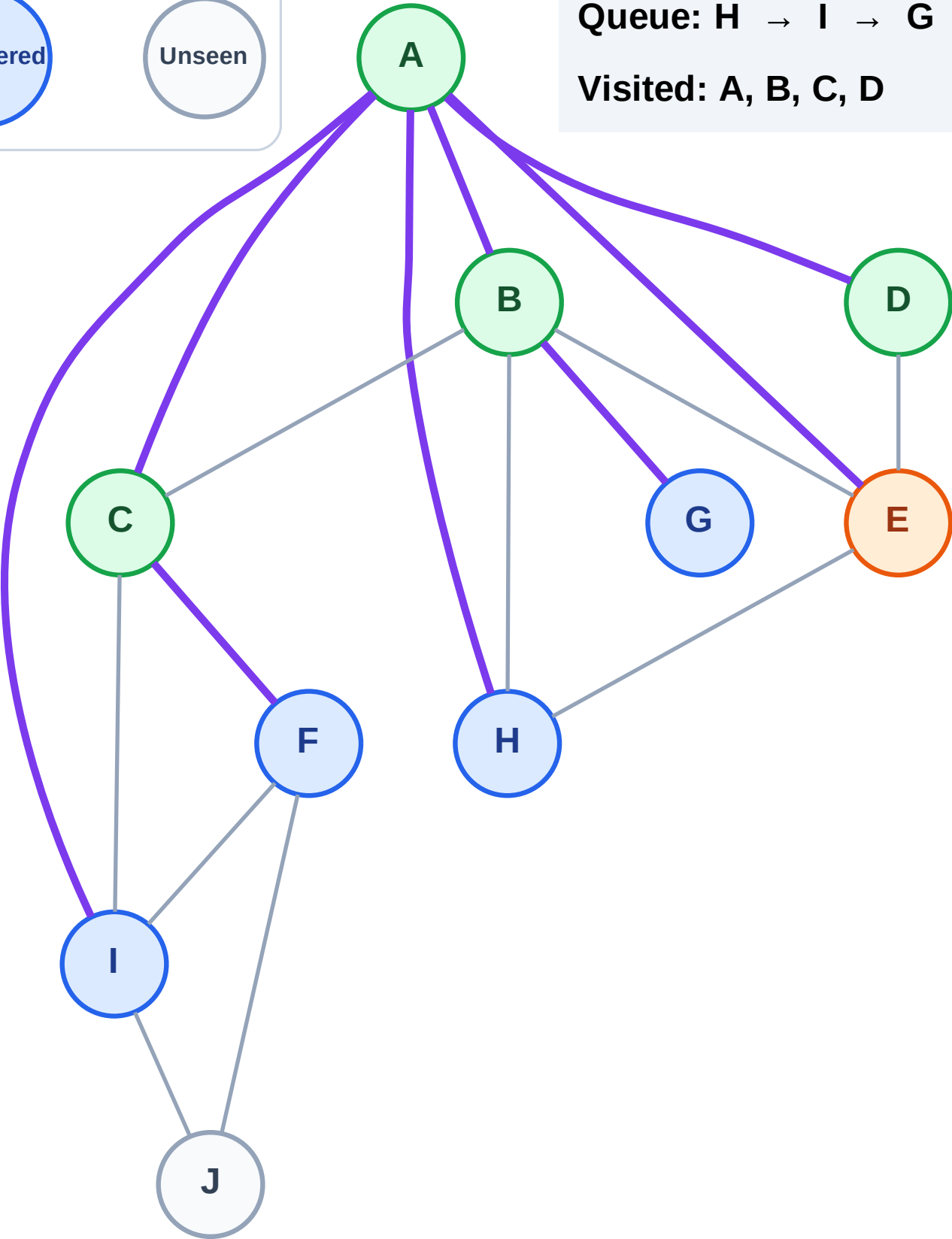
Processing

Discovered

Unseen

Queue: H → I → G → F

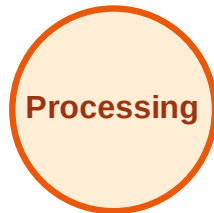
Visited: A, B, C, D



BFS Traversal

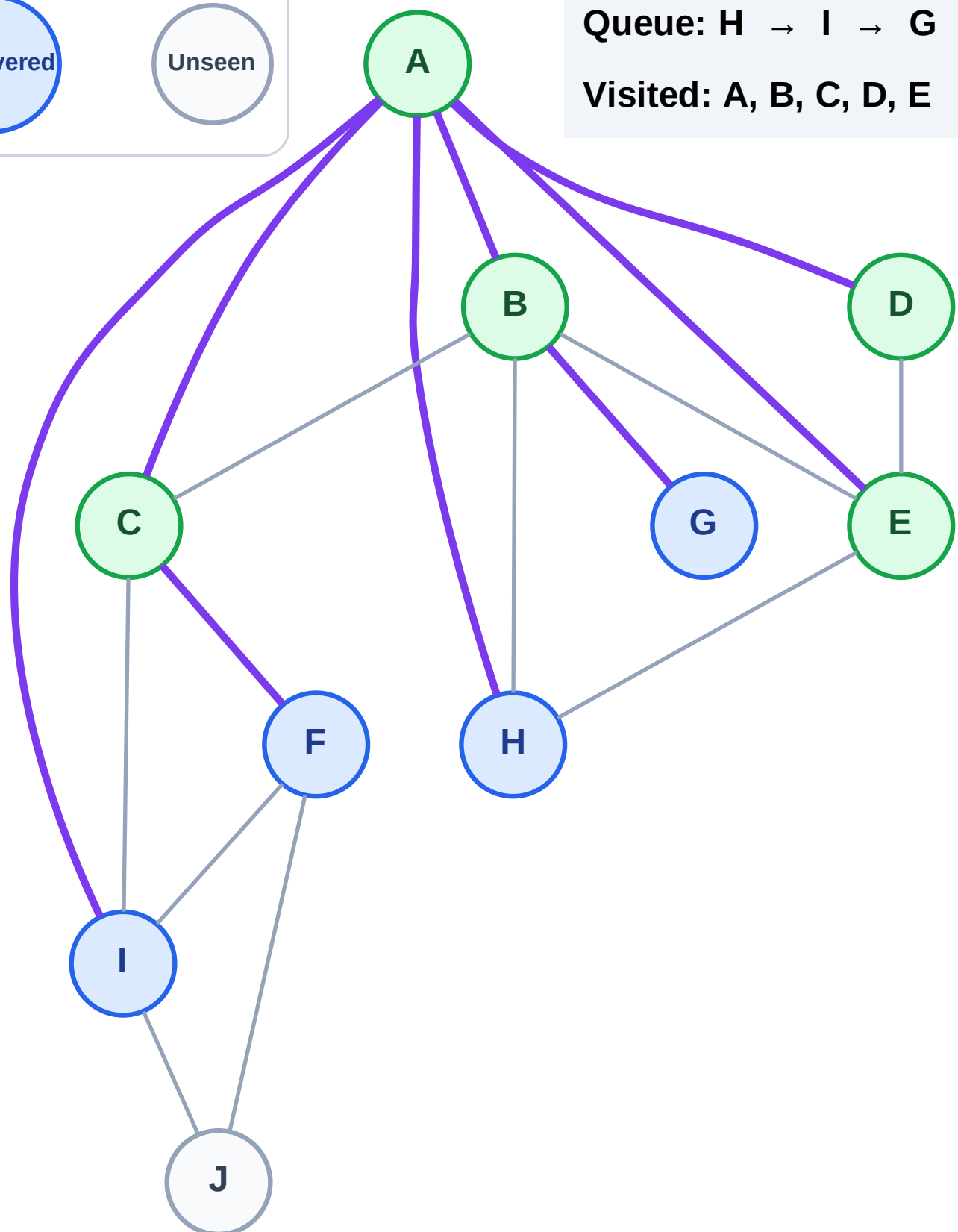
Step 11: No new neighbors from E

Legend



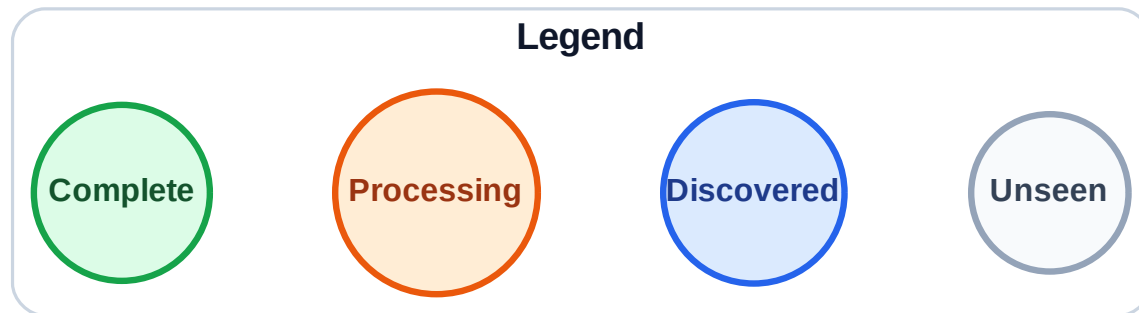
Queue: H → I → G → F

Visited: A, B, C, D, E

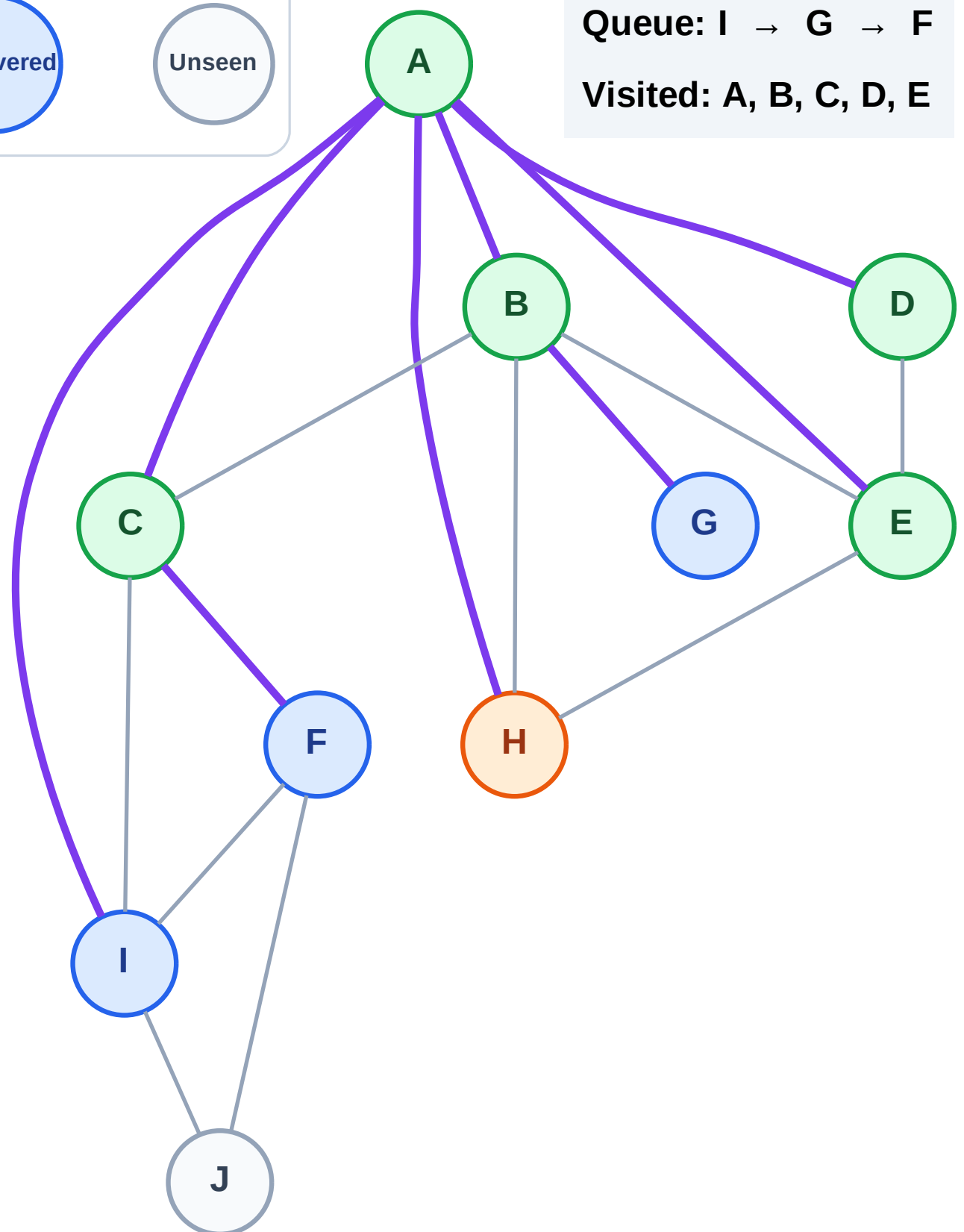


BFS Traversal

Step 12: Process node H



Queue: I → G → F
Visited: A, B, C, D, E



BFS Traversal

Step 13: No new neighbors from H

Legend

Complete

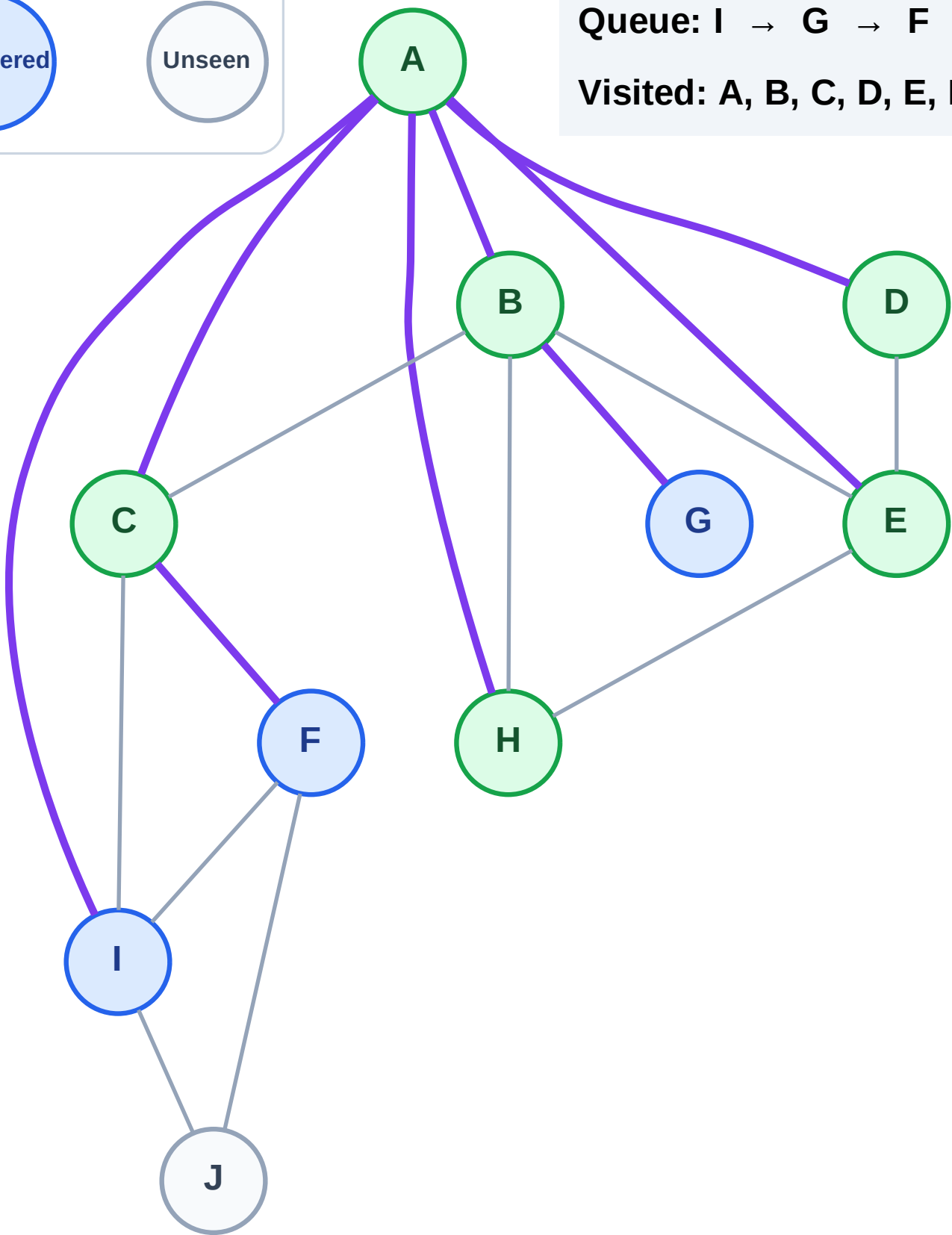
Processing

Discovered

Unseen

Queue: I → G → F

Visited: A, B, C, D, E, H



BFS Traversal

Step 14: Process node I

Legend

Complete

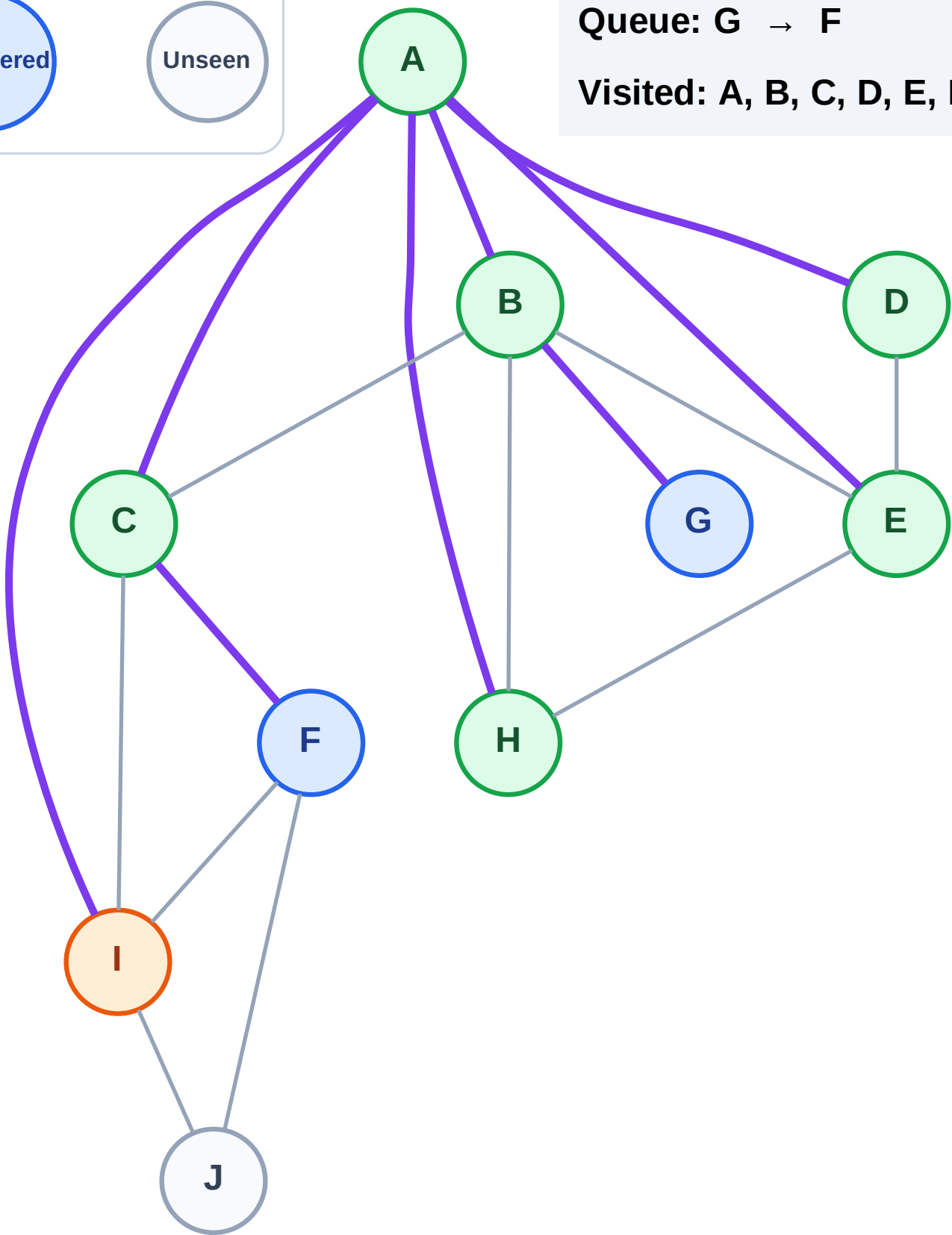
Processing

Discovered

Unseen

Queue: G → F

Visited: A, B, C, D, E, H



BFS Traversal

Step 15: Discover J from I

Legend

Complete

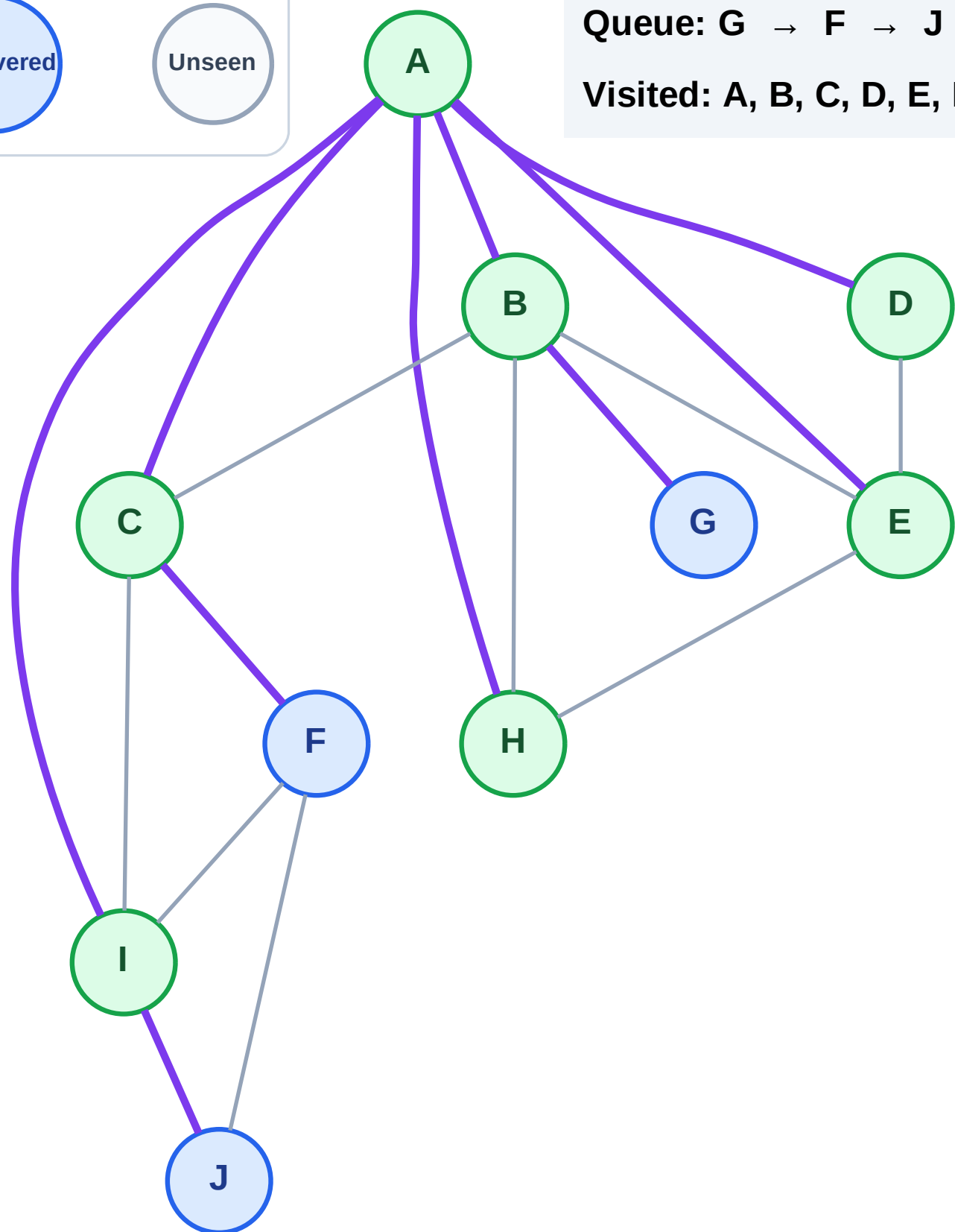
Processing

Discovered

Unseen

Queue: G → F → J

Visited: A, B, C, D, E, H, I



BFS Traversal

Step 16: Process node G

Legend

Complete

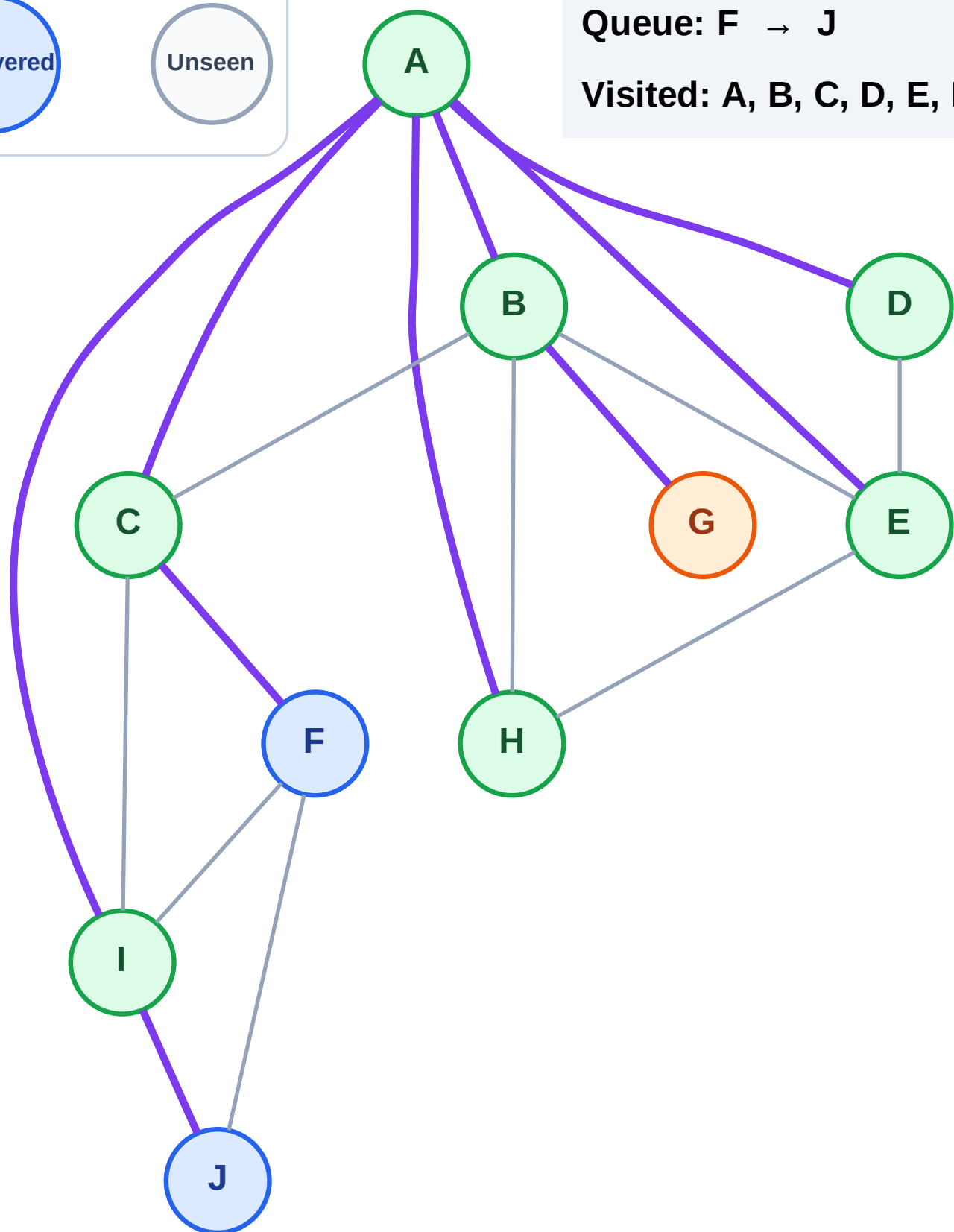
Processing

Discovered

Unseen

Queue: F → J

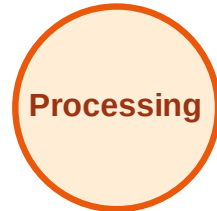
Visited: A, B, C, D, E, H, I



BFS Traversal

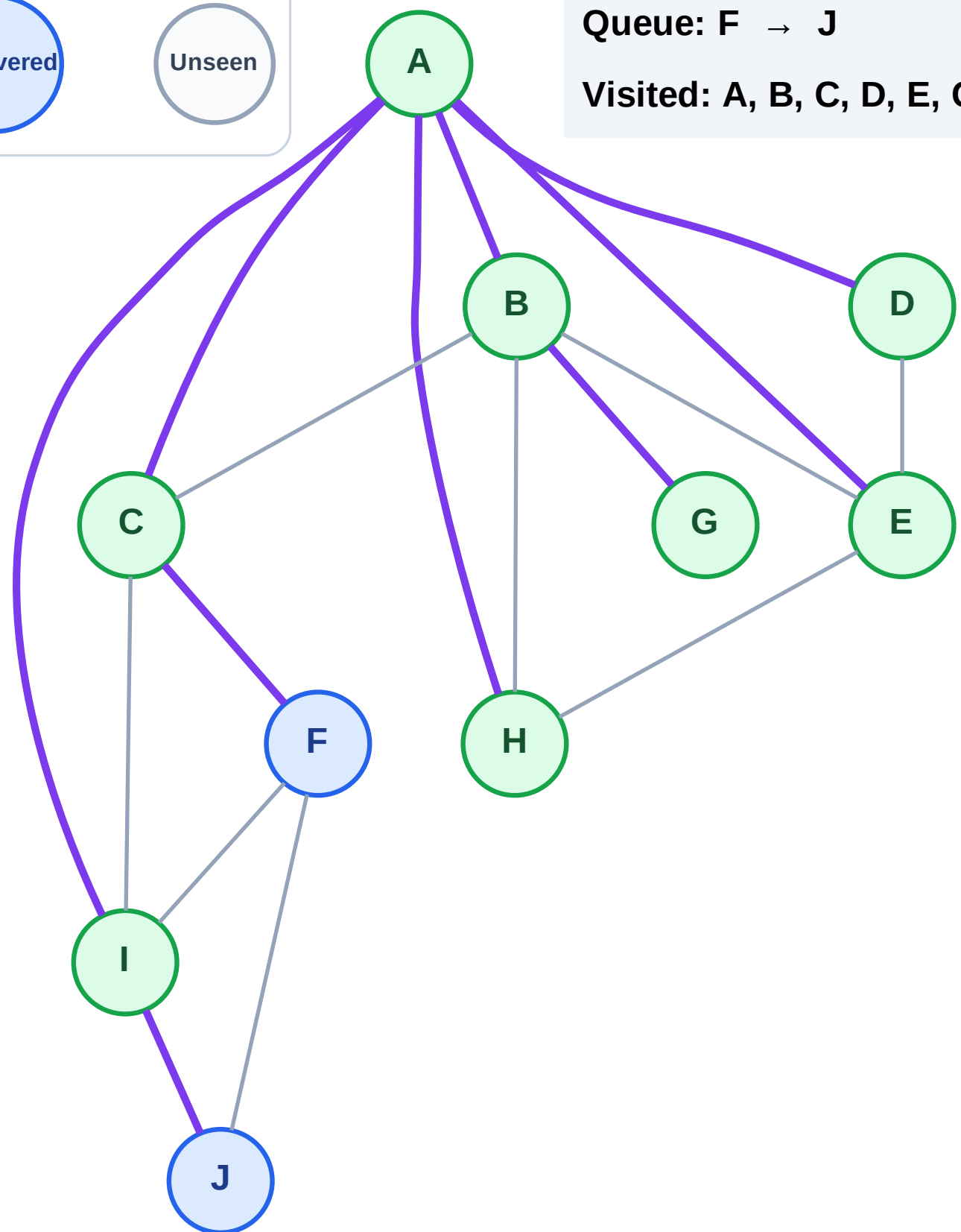
Step 17: No new neighbors from G

Legend



Queue: F → J

Visited: A, B, C, D, E, G, H, I



BFS Traversal

Step 18: Process node F

Legend

Complete

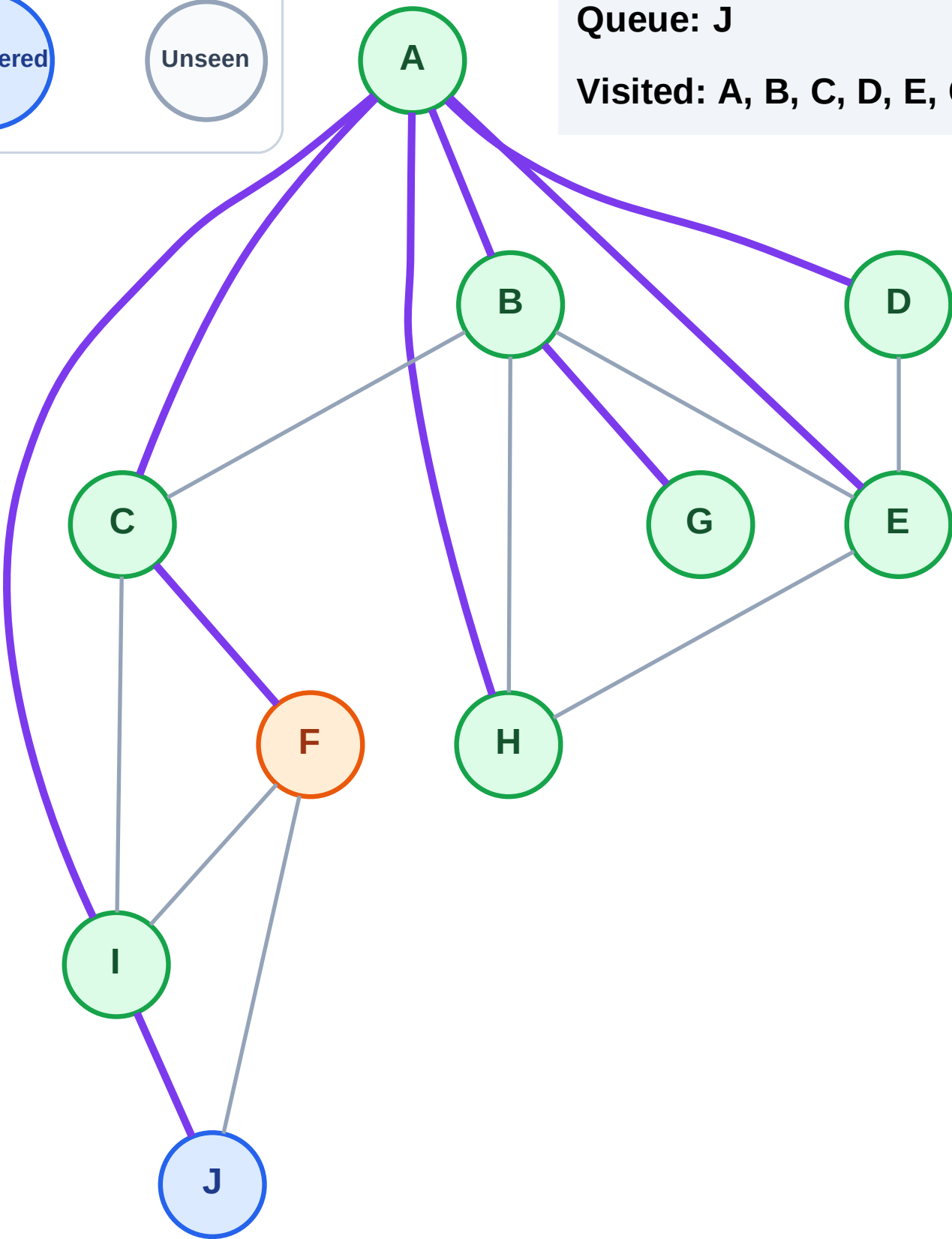
Processing

Discovered

Unseen

Queue: J

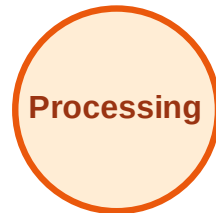
Visited: A, B, C, D, E, G, H, I



BFS Traversal

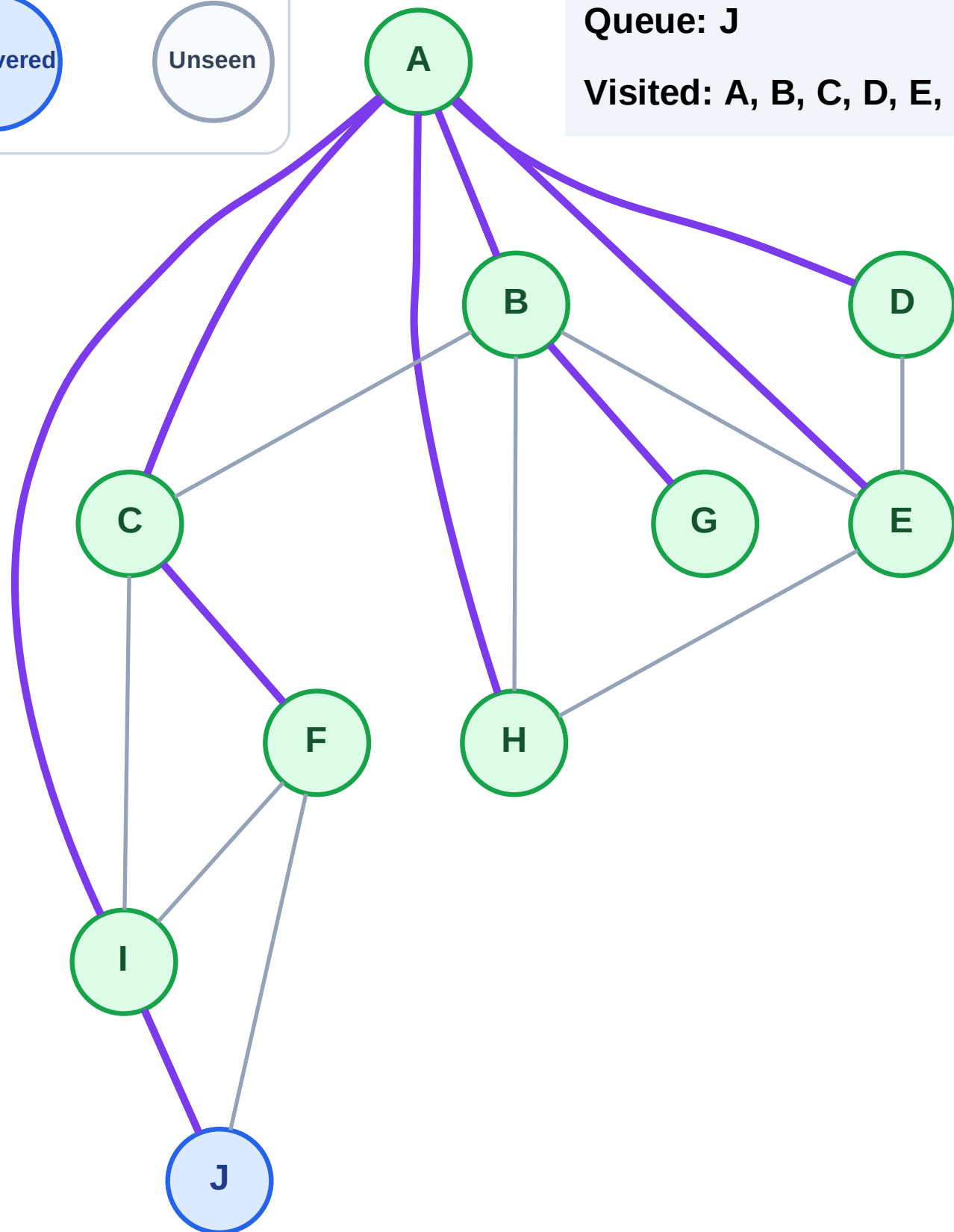
Step 19: No new neighbors from F

Legend



Queue: J

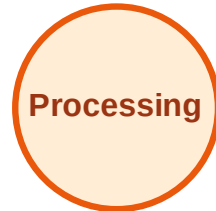
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

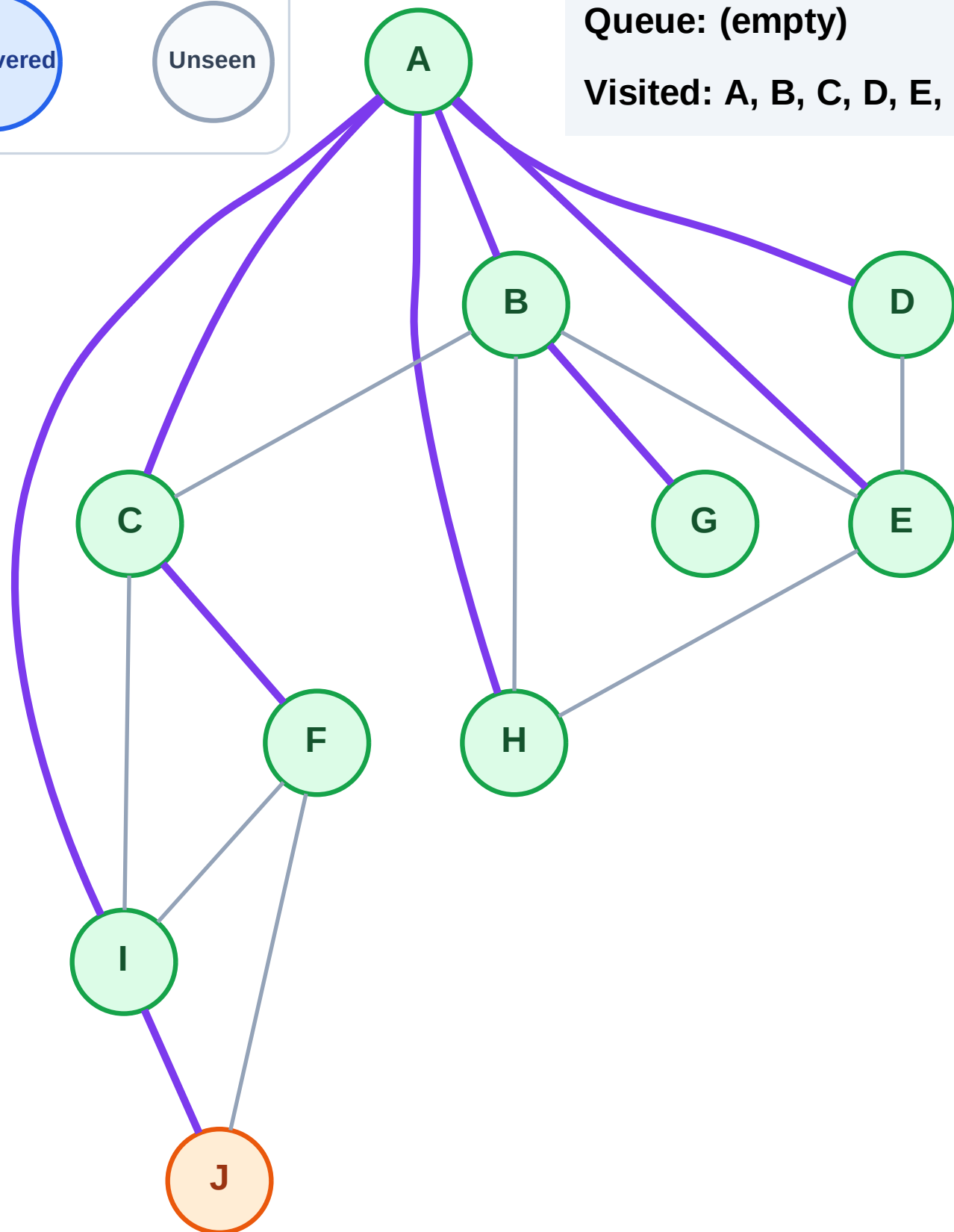
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

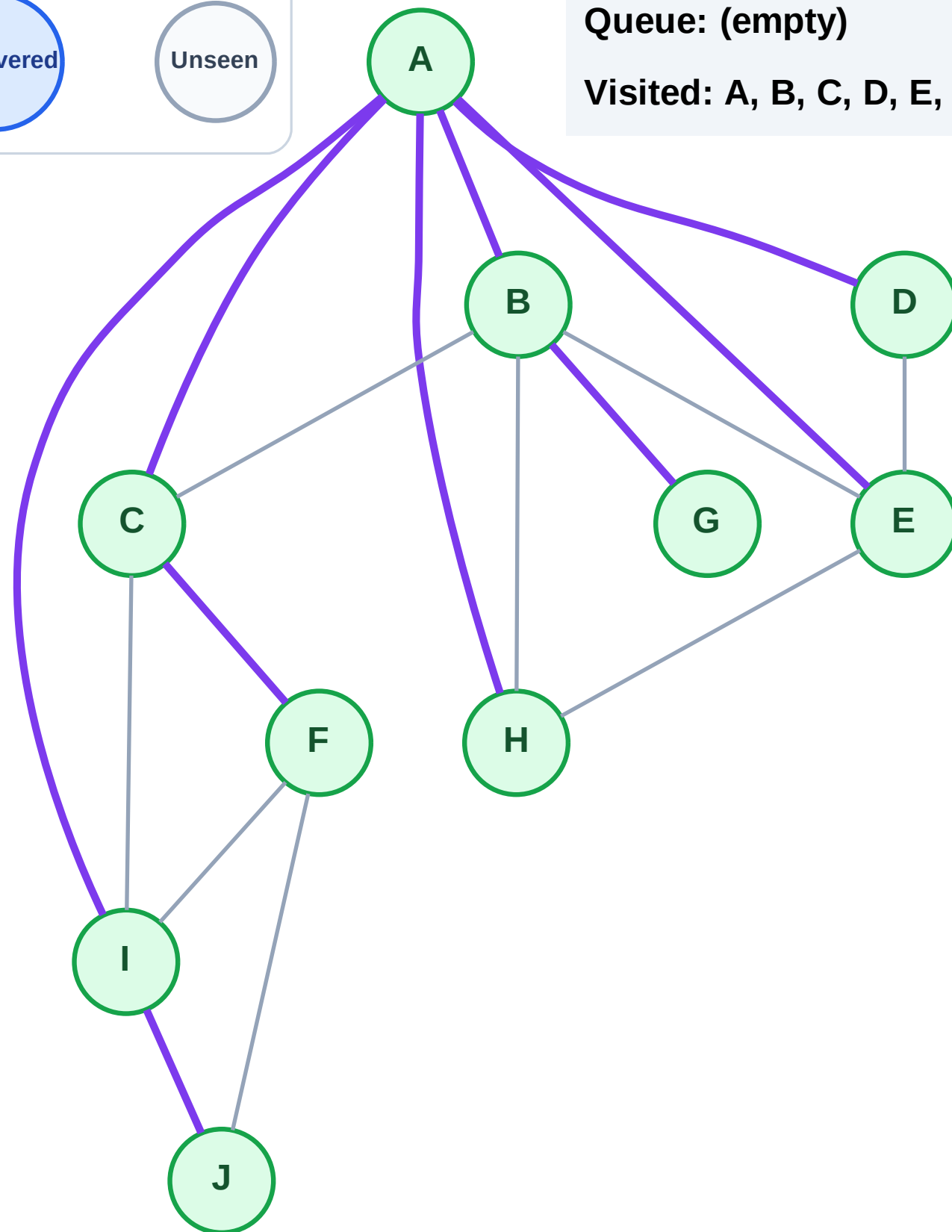
Step 21: No new neighbors from J

Legend



Queue: (empty)

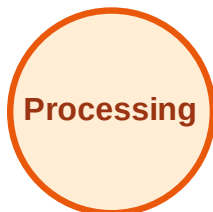
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

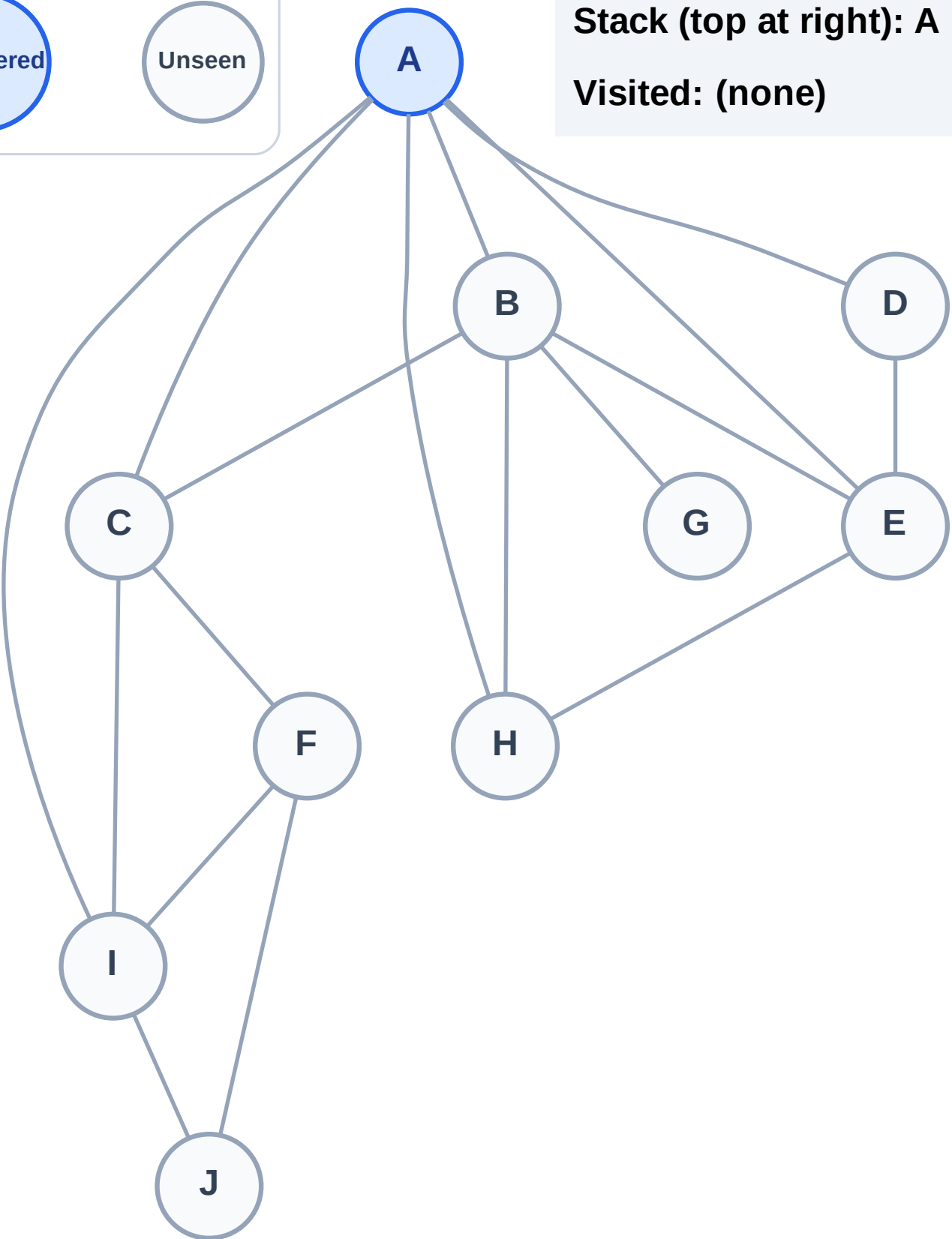
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

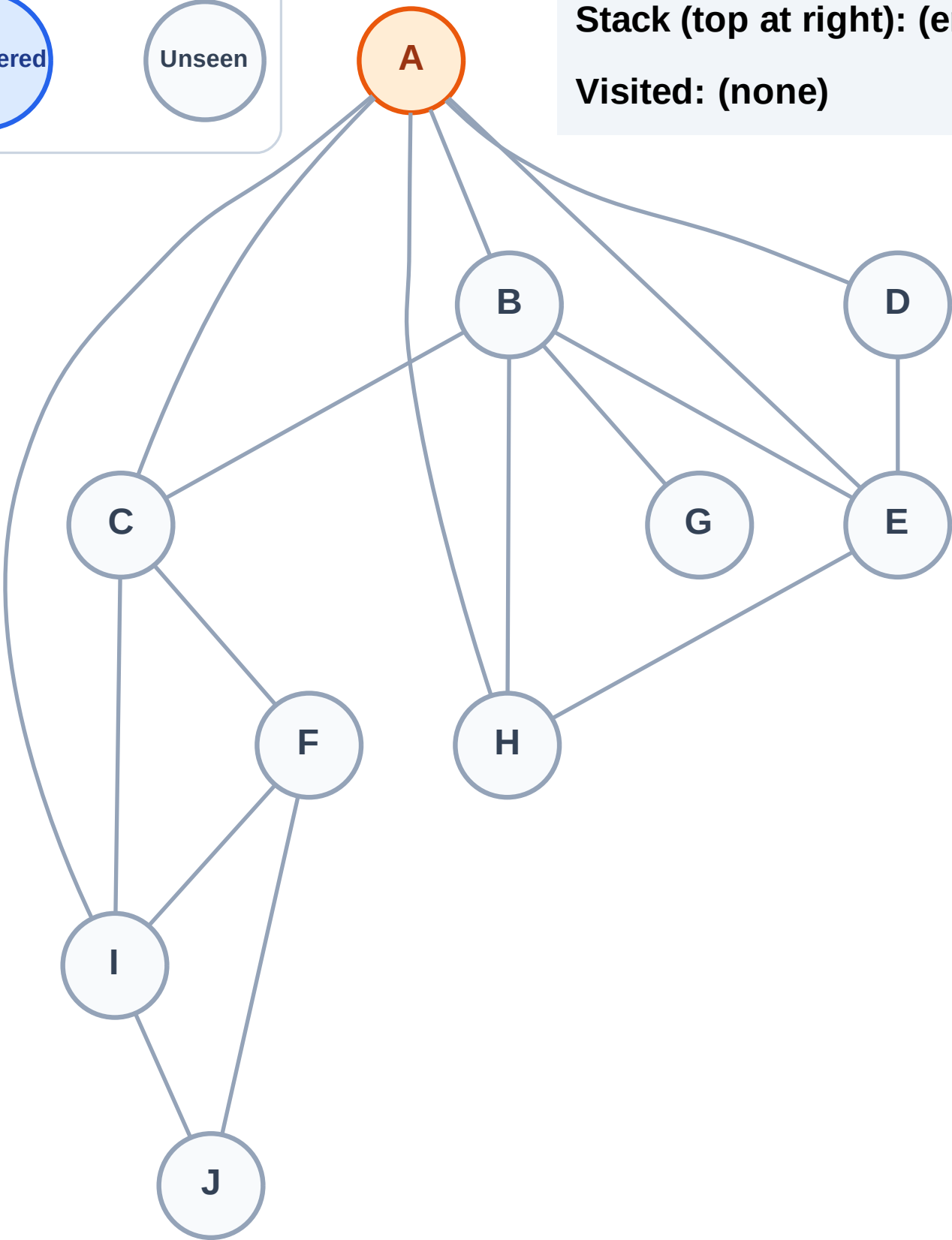
Complete

Processing

Discovered

Unseen

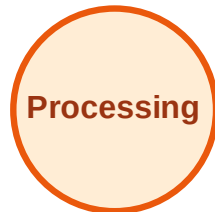
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

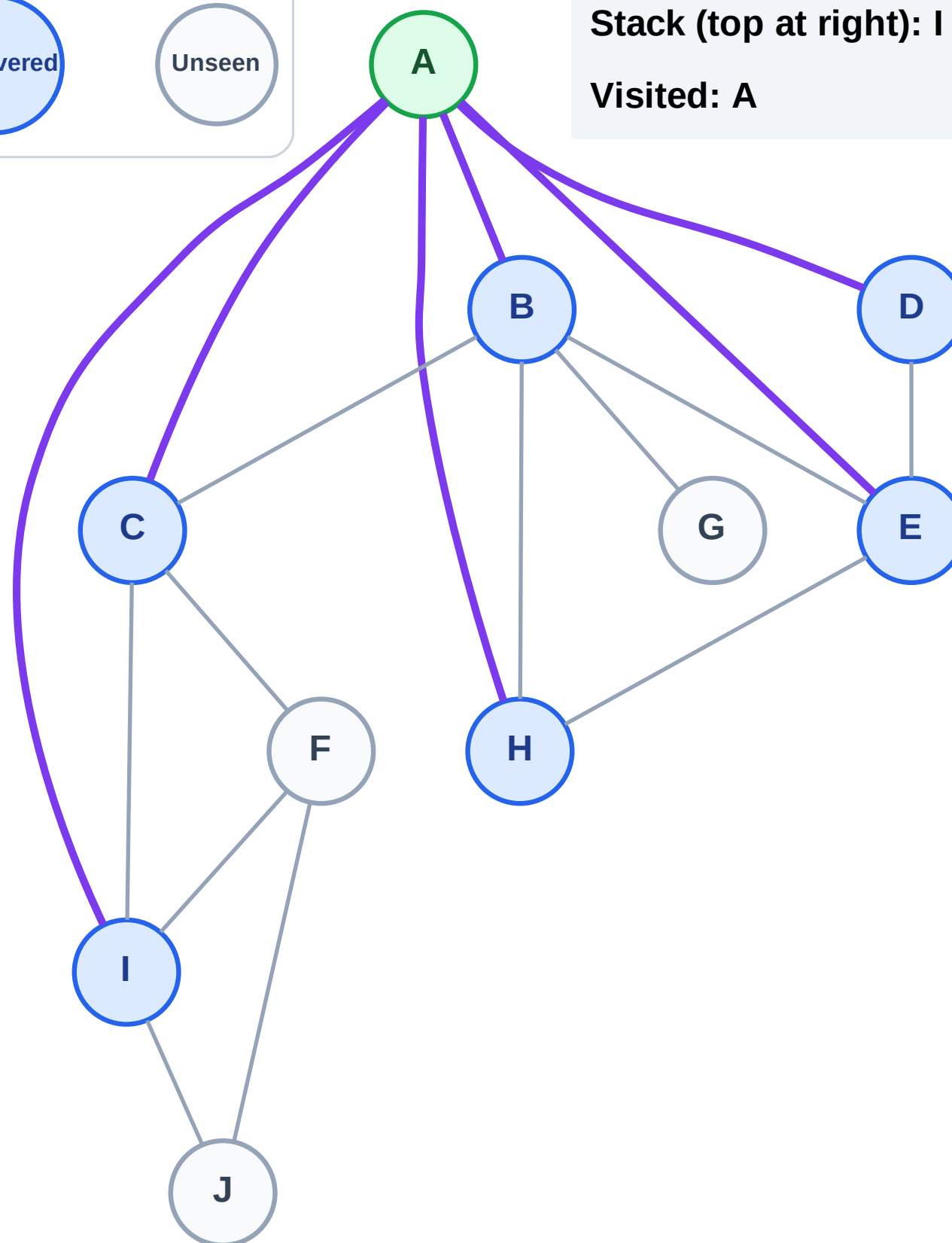
Step 3: Discover I, H, E, D, C, B from A

Legend



Stack (top at right): I | H | E | D | C | B

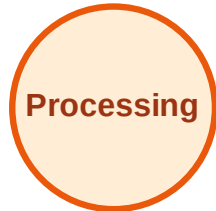
Visited: A



DFS Traversal

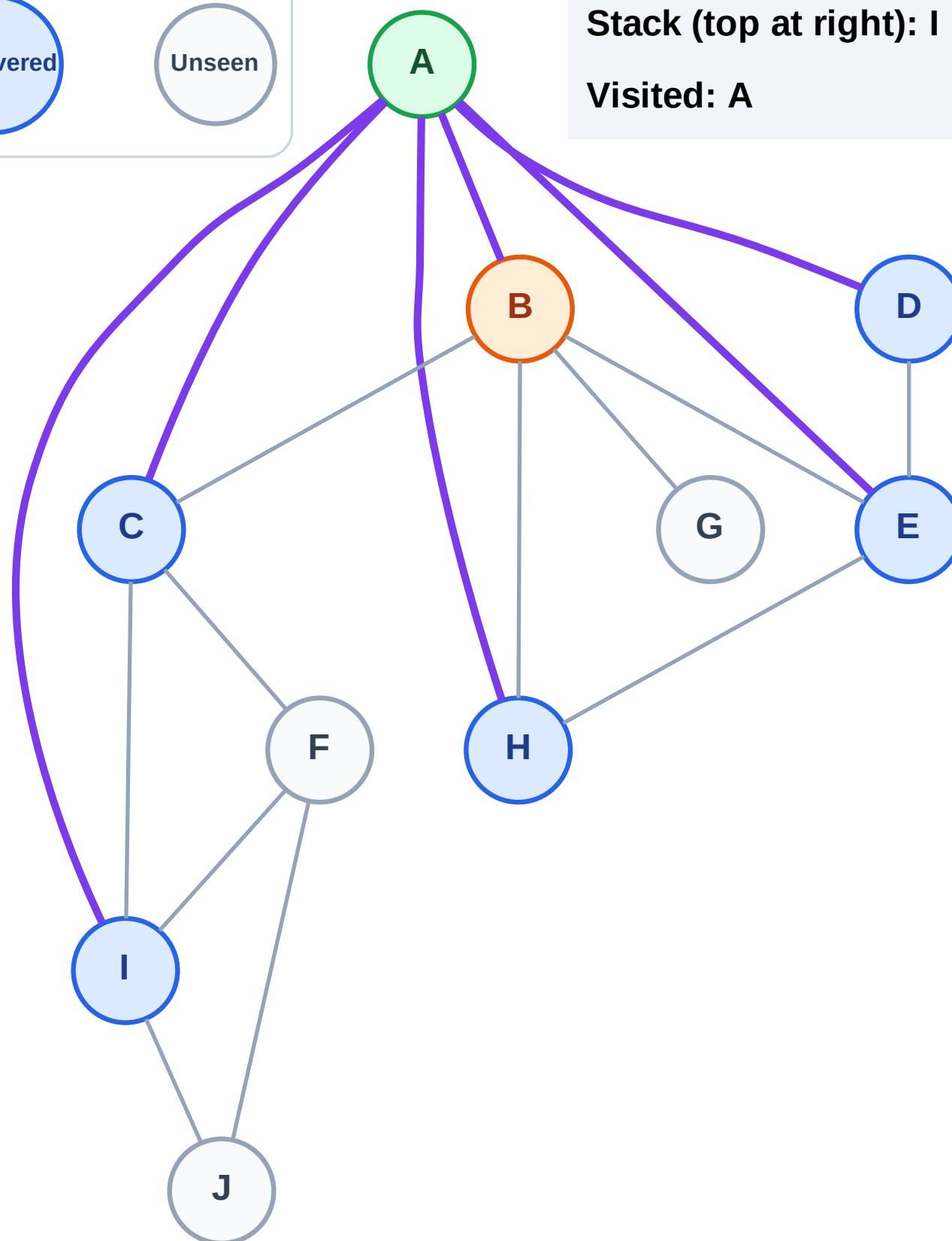
Step 4: Process node B

Legend



Stack (top at right): I | H | E | D | C

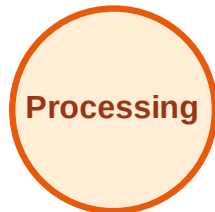
Visited: A



DFS Traversal

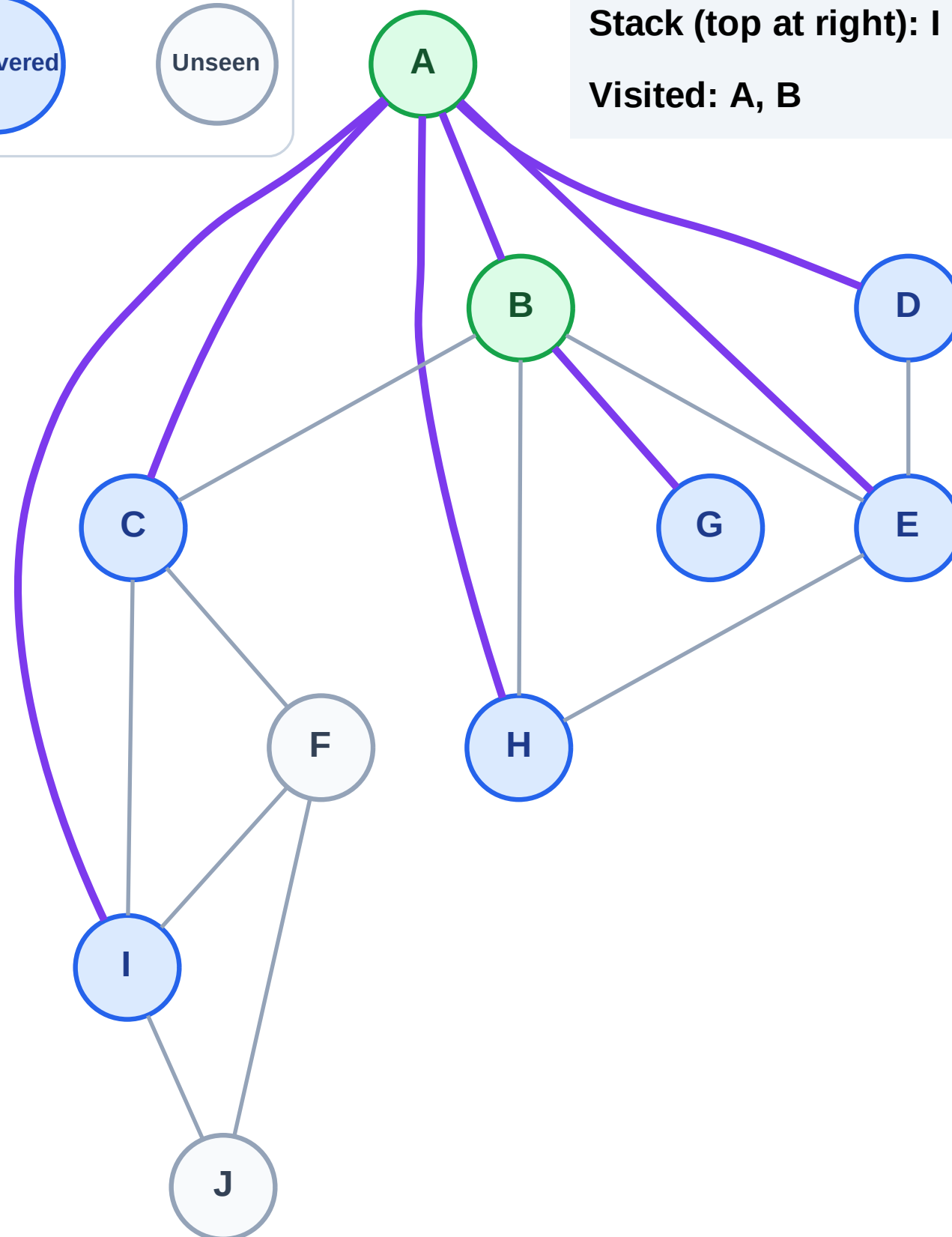
Step 5: Discover G from B

Legend



Stack (top at right): I | H | E | D | C | G

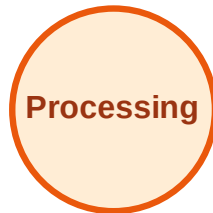
Visited: A, B



DFS Traversal

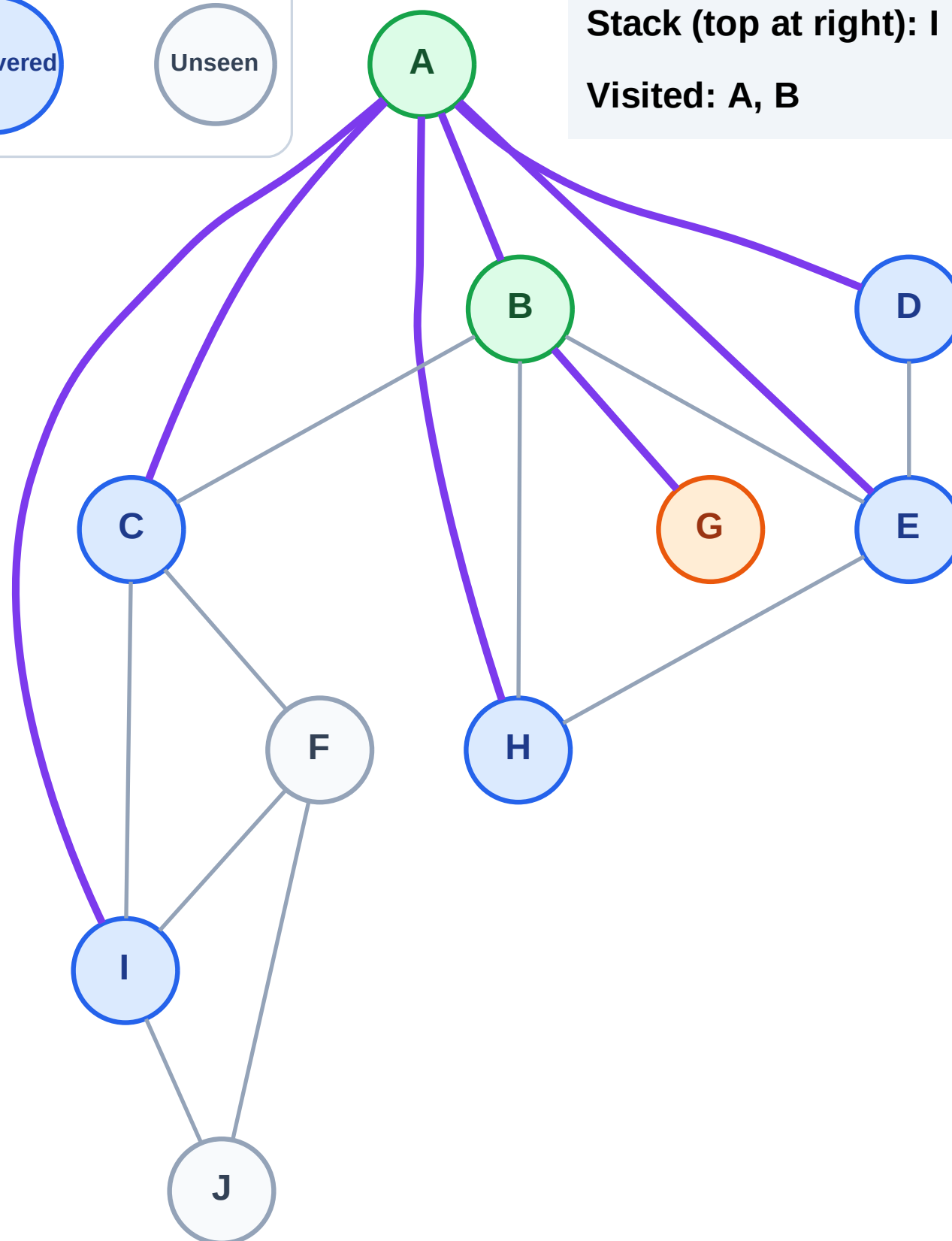
Step 6: Process node G

Legend



Stack (top at right): I | H | E | D | C

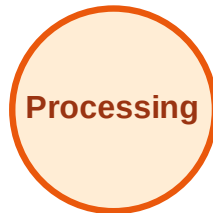
Visited: A, B



DFS Traversal

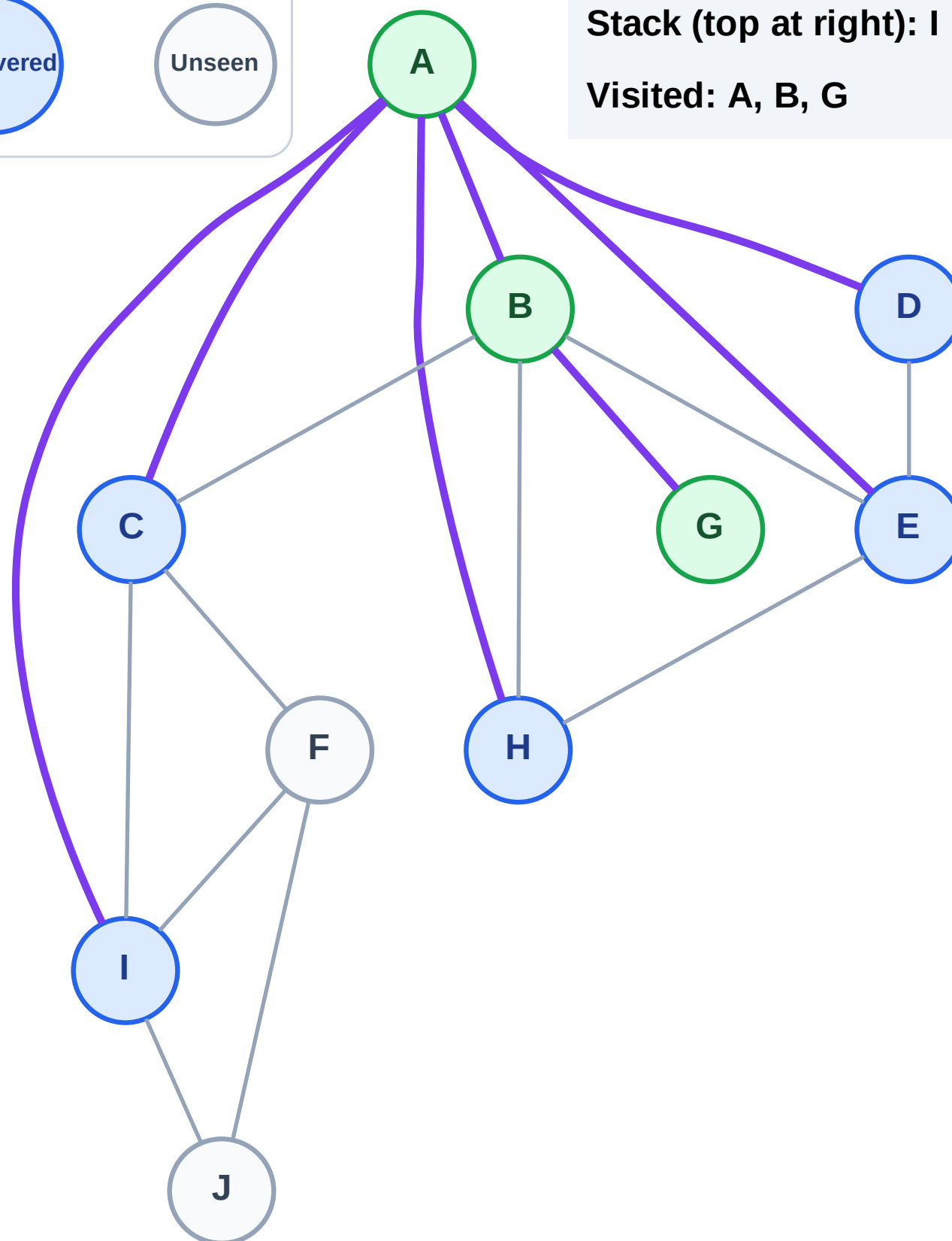
Step 7: No new neighbors from G

Legend



Stack (top at right): I | H | E | D | C

Visited: A, B, G



DFS Traversal

Step 8: Process node C

Legend

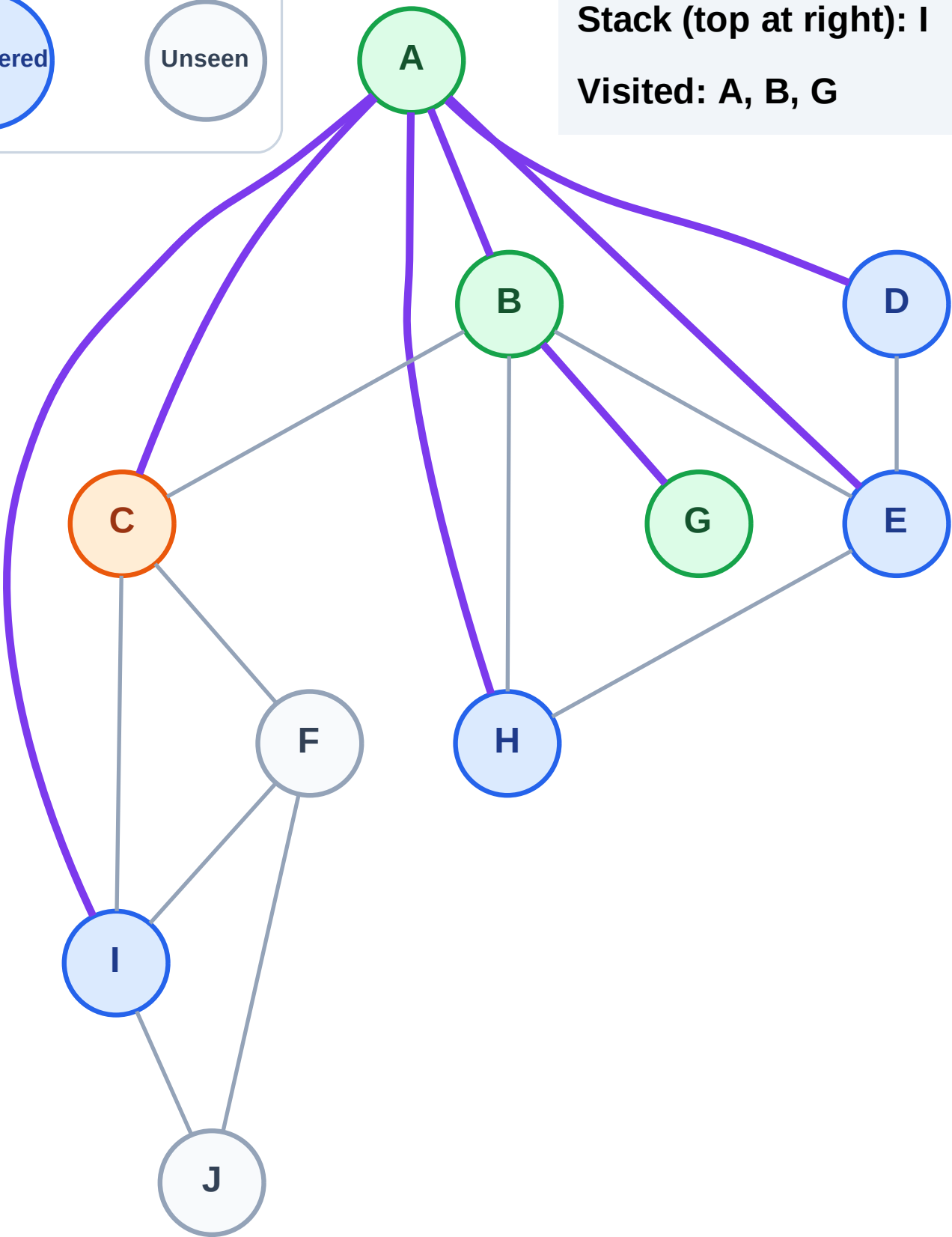
Complete

Processing

Discovered

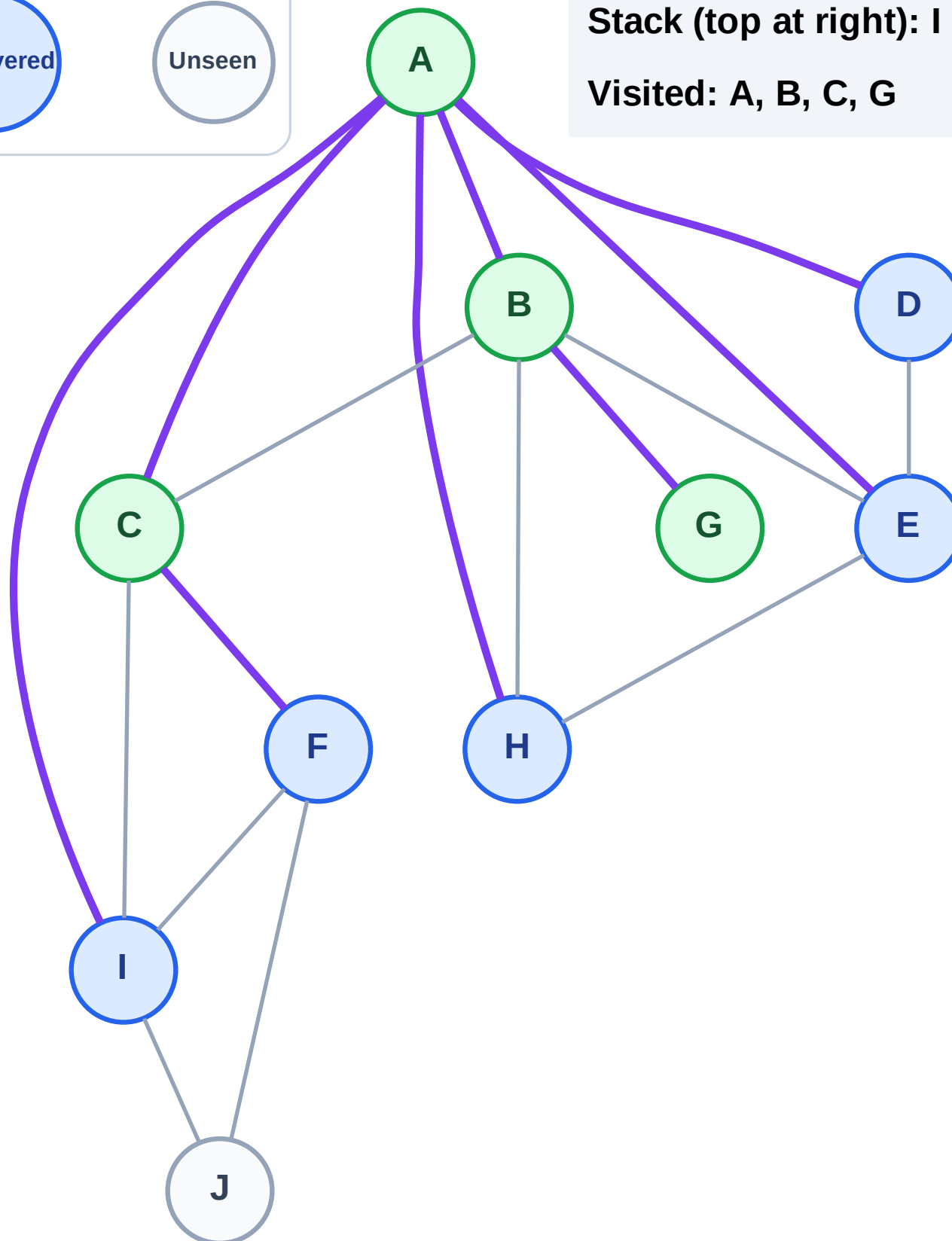
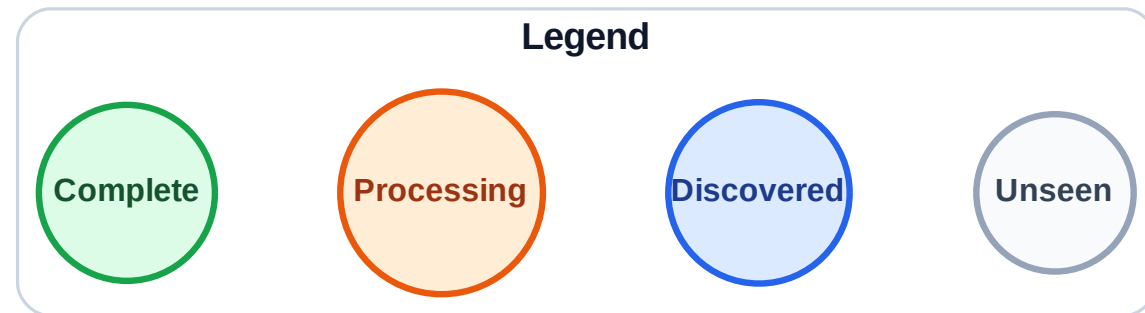
Unseen

Stack (top at right): I | H | E | D
Visited: A, B, G



Step 9: Discover F from C

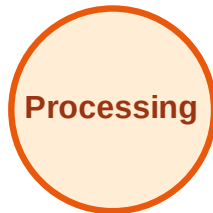
Step 9: Discover F from C



DFS Traversal

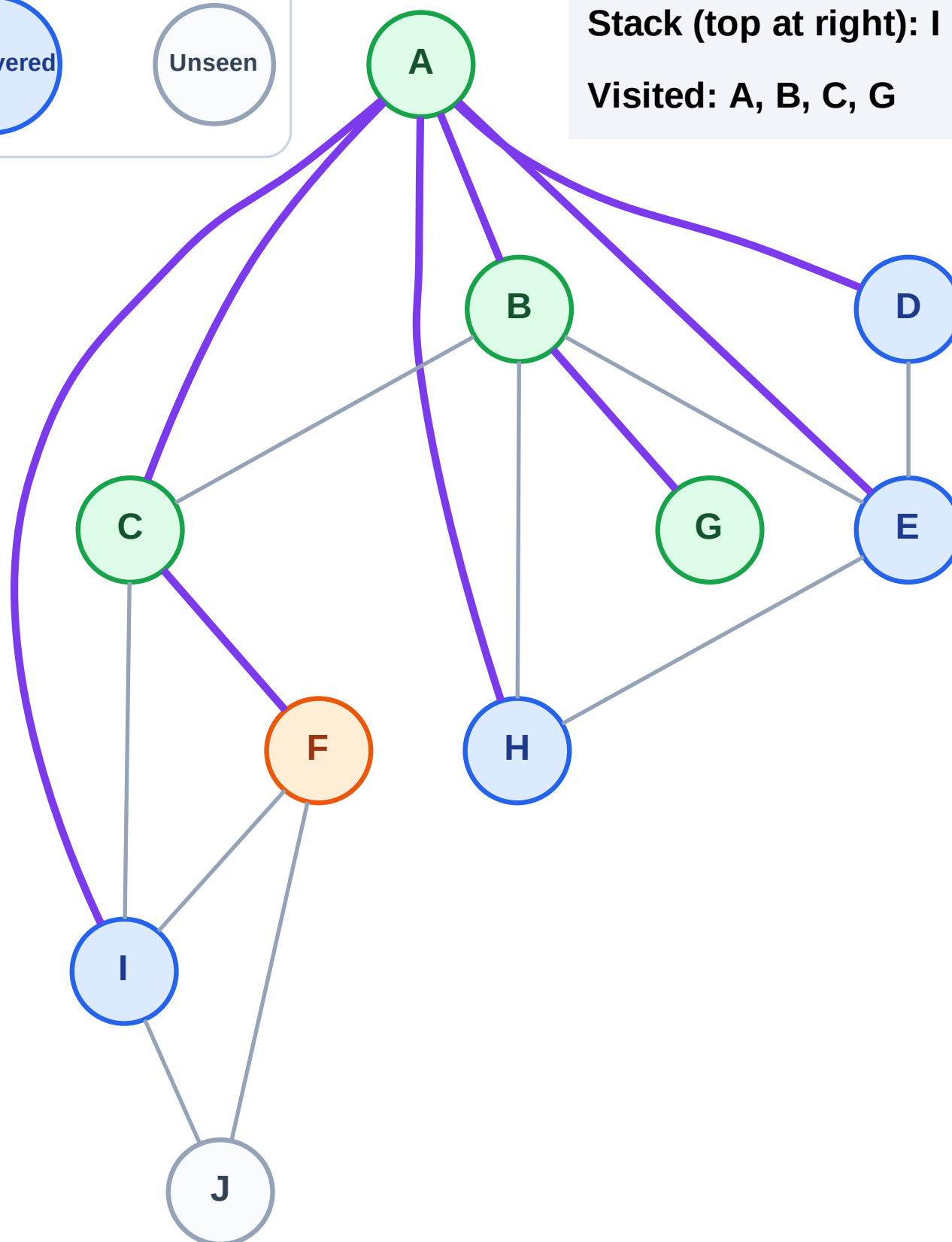
Step 10: Process node F

Legend



Stack (top at right): I | H | E | D

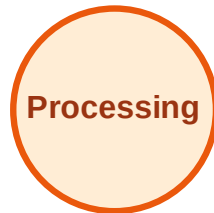
Visited: A, B, C, G



DFS Traversal

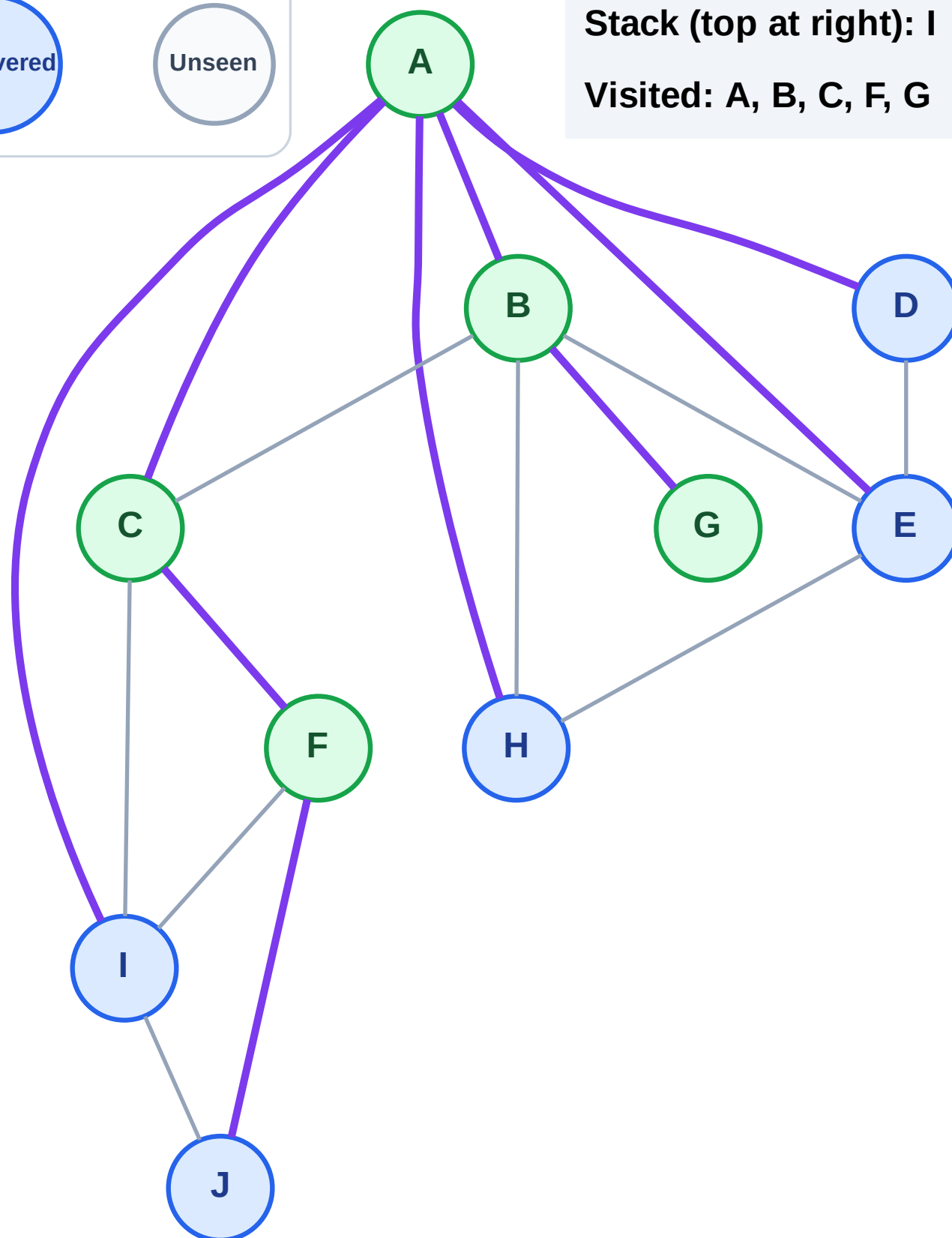
Step 11: Discover J from F

Legend



Stack (top at right): I | H | E | D | J

Visited: A, B, C, F, G



DFS Traversal

Step 12: Process node J

Legend

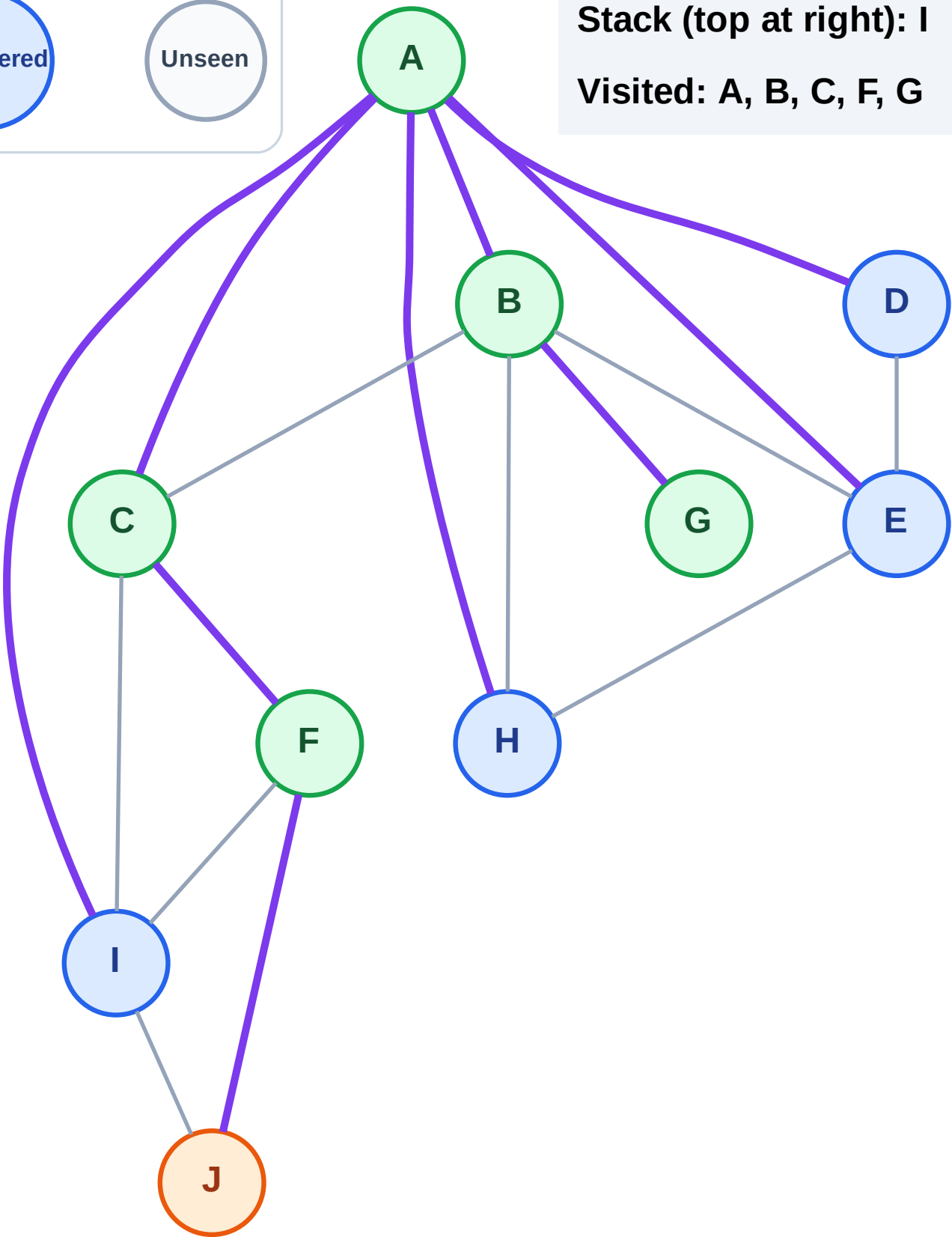
Complete

Processing

Discovered

Unseen

Stack (top at right): I | H | E | D
Visited: A, B, C, F, G



DFS Traversal

Step 13: No new neighbors from J

Legend

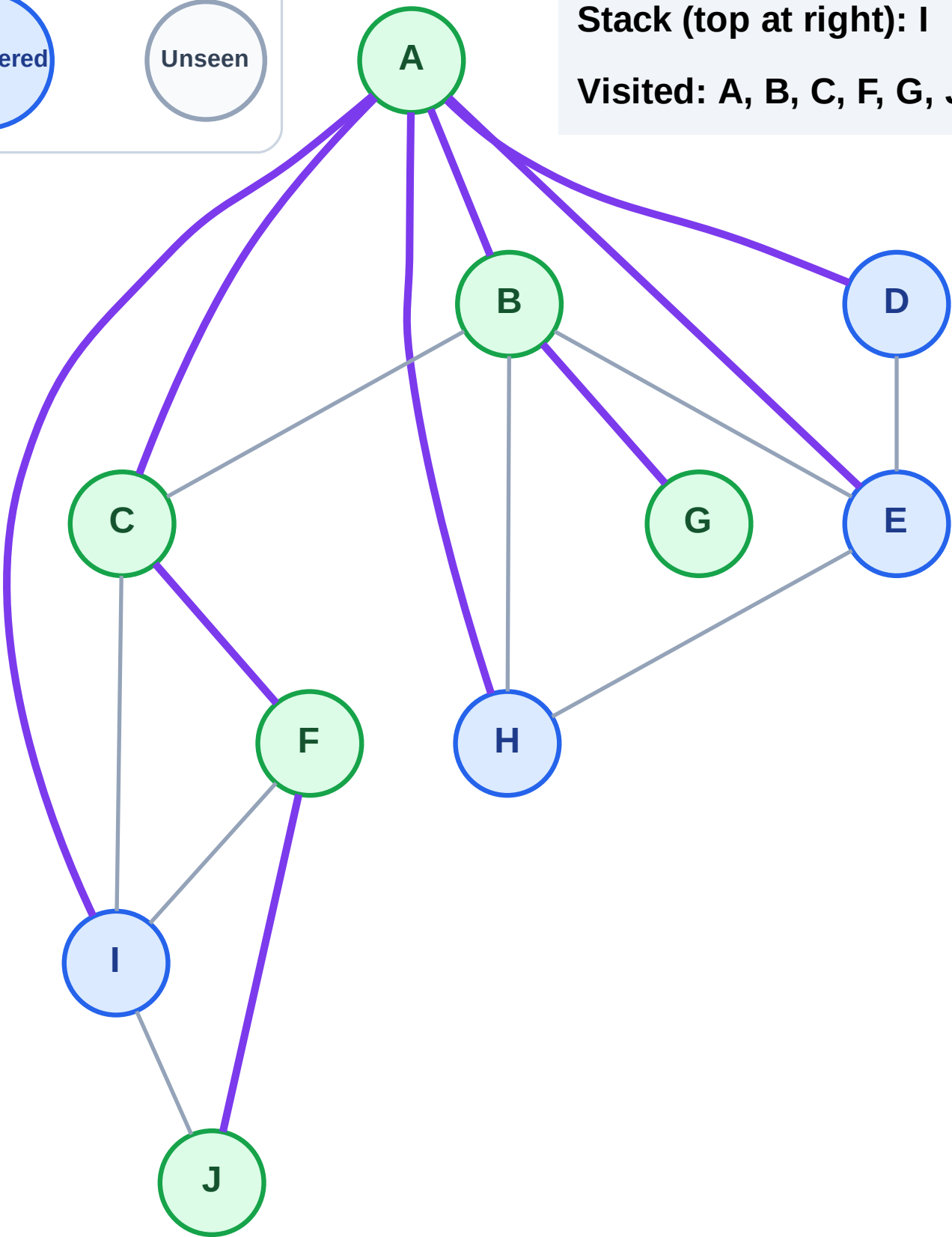
Complete

Processing

Discovered

Unseen

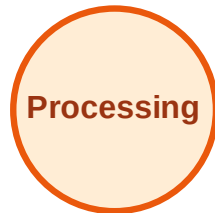
Stack (top at right): I | H | E | D
Visited: A, B, C, F, G, J



DFS Traversal

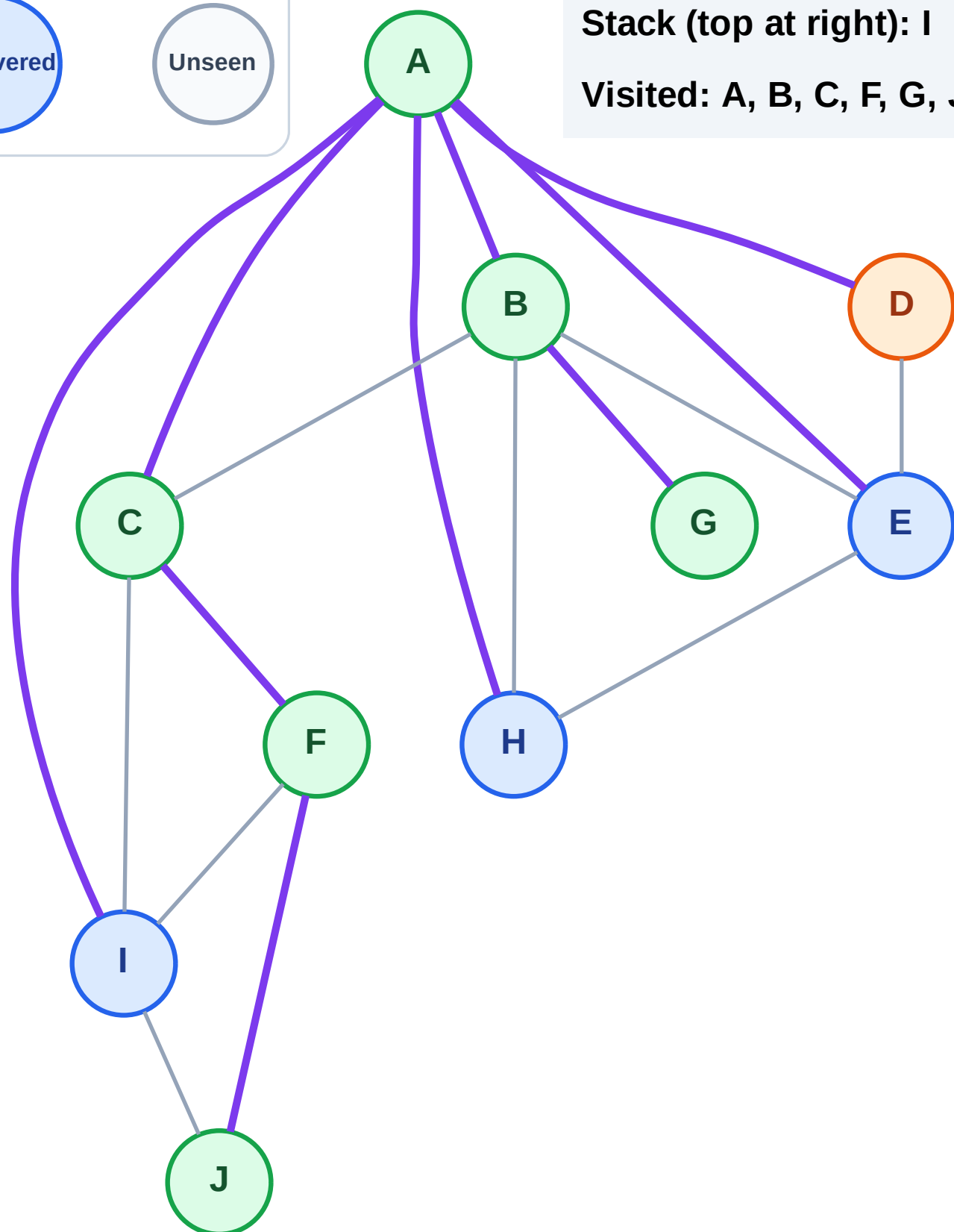
Step 14: Process node D

Legend



Stack (top at right): I | H | E

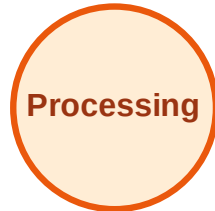
Visited: A, B, C, F, G, J



DFS Traversal

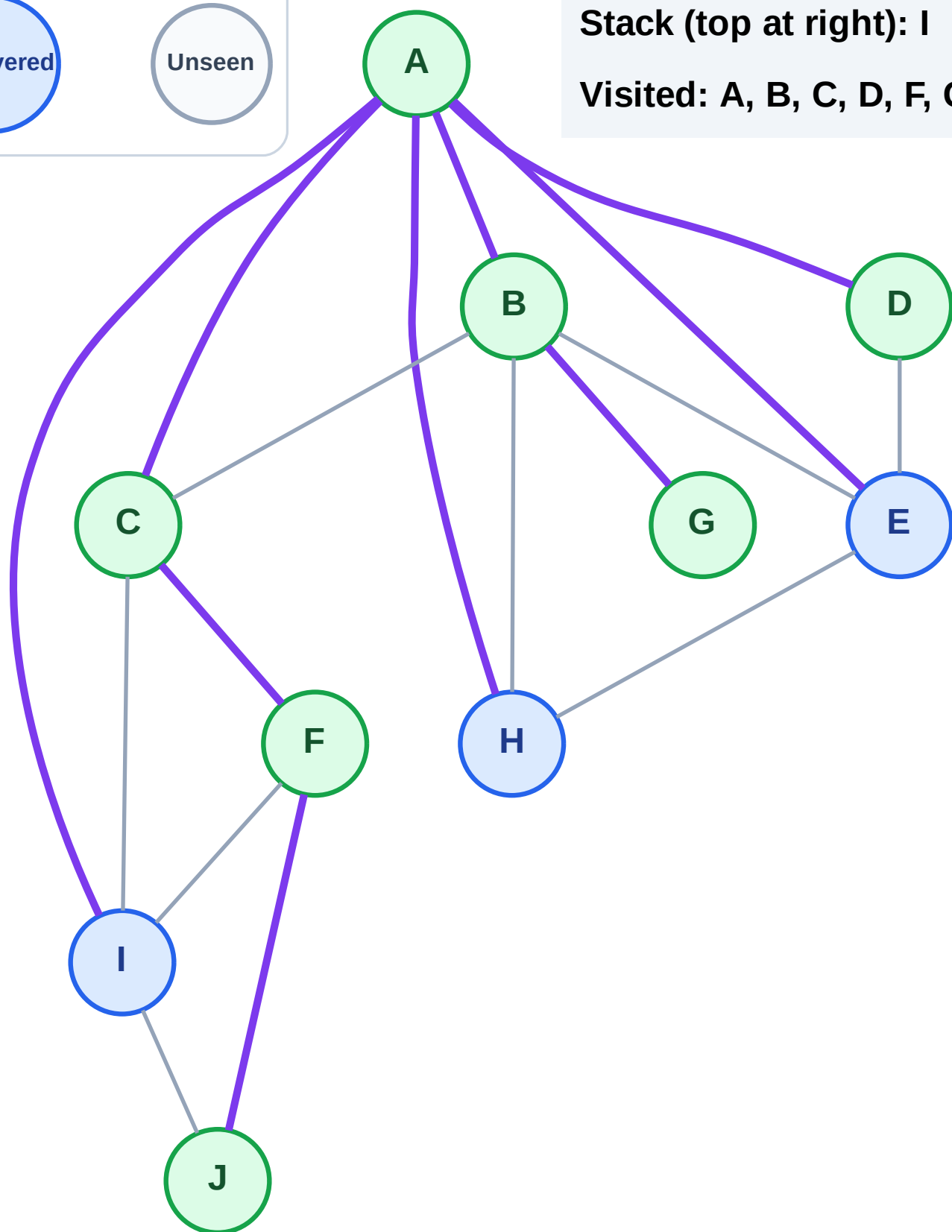
Step 15: No new neighbors from D

Legend



Stack (top at right): I | H | E

Visited: A, B, C, D, F, G, J



DFS Traversal

Step 16: Process node E

Legend

Complete

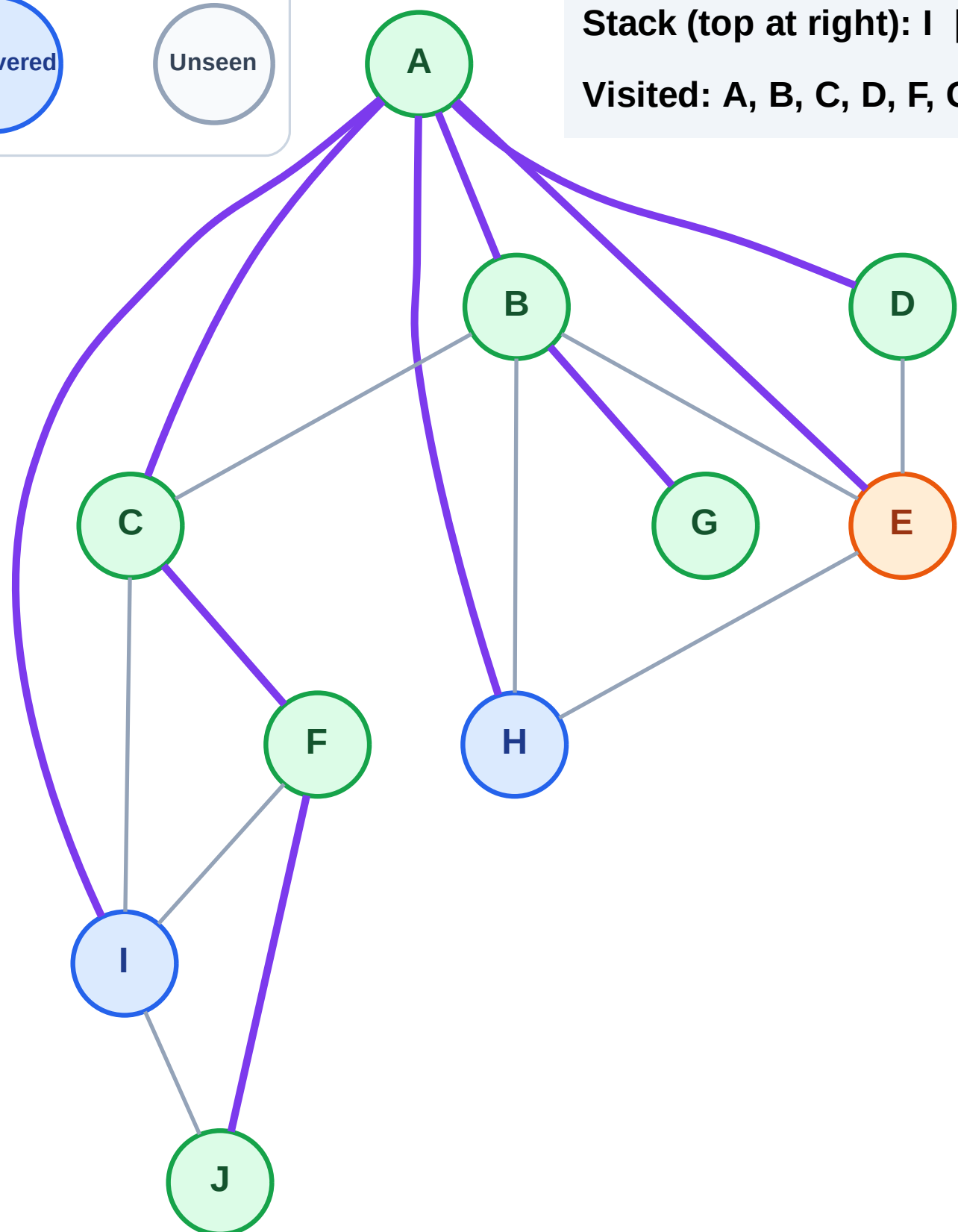
Processing

Discovered

Unseen

Stack (top at right): I | H

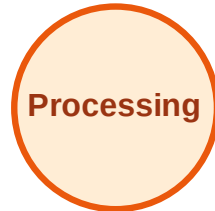
Visited: A, B, C, D, F, G, J



DFS Traversal

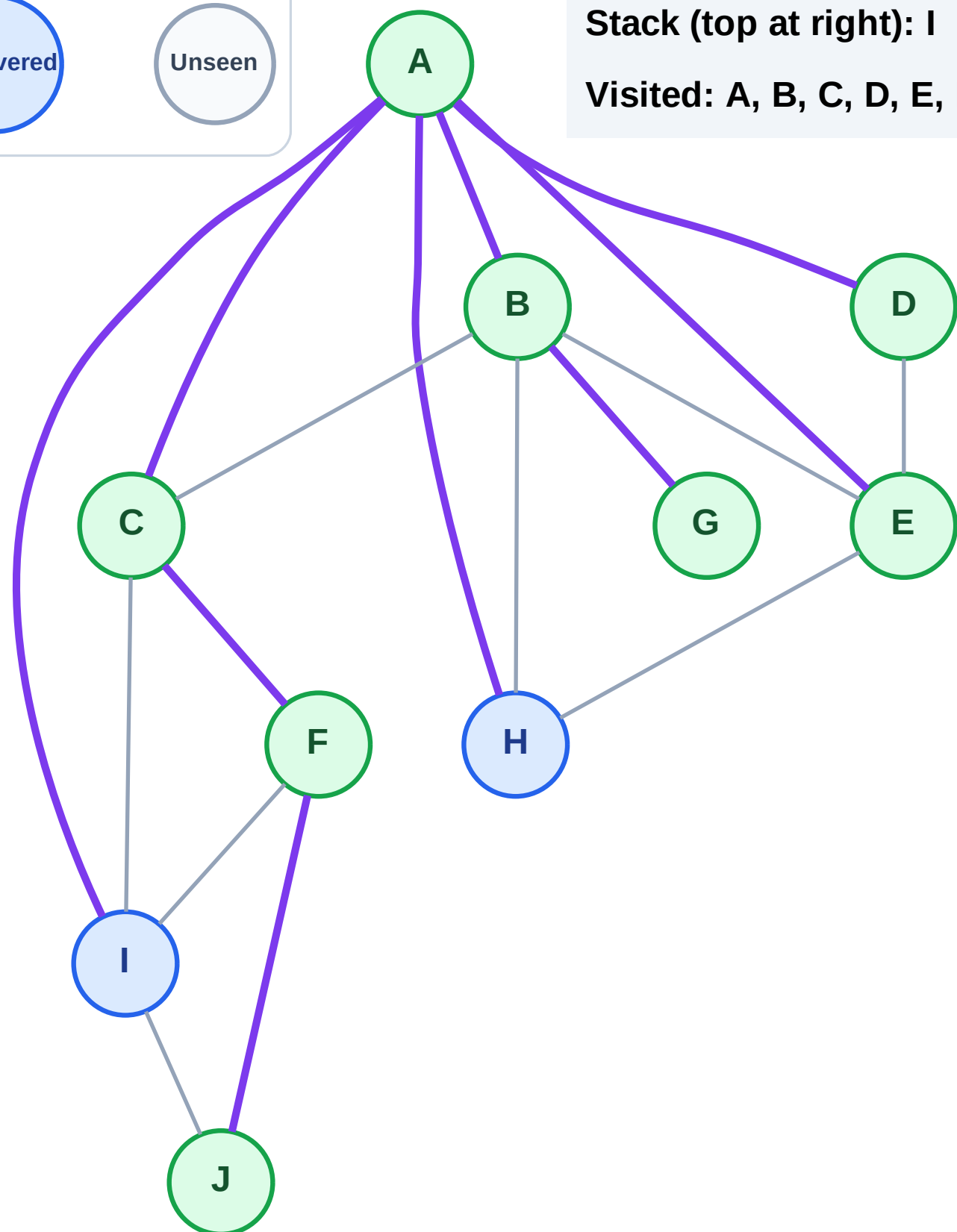
Step 17: No new neighbors from E

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



DFS Traversal

Step 18: Process node H

Legend

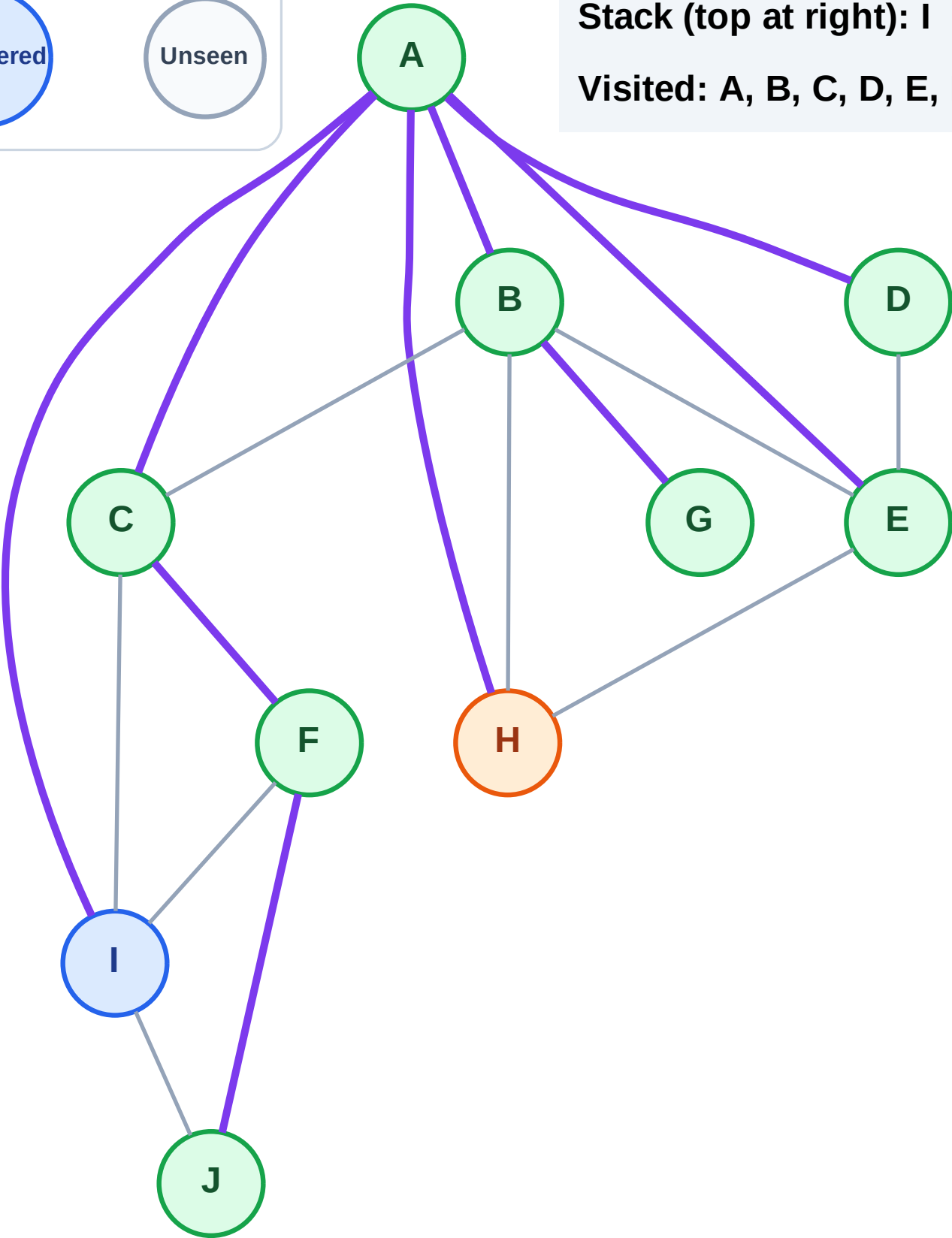
Complete

Processing

Discovered

Unseen

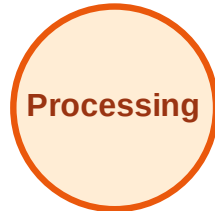
Stack (top at right): I
Visited: A, B, C, D, E, F, G, J



DFS Traversal

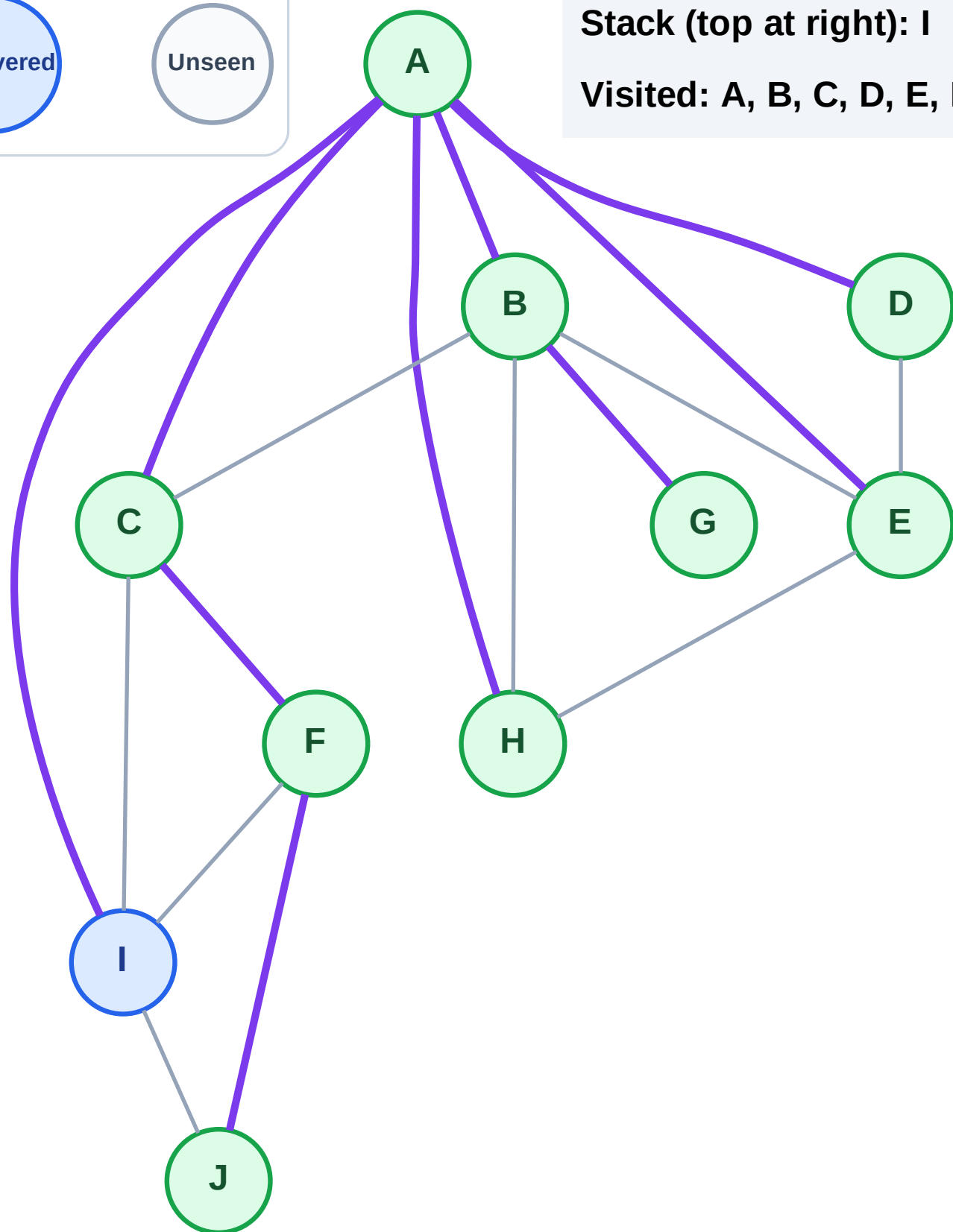
Step 19: No new neighbors from H

Legend



Stack (top at right): I

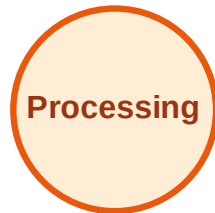
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

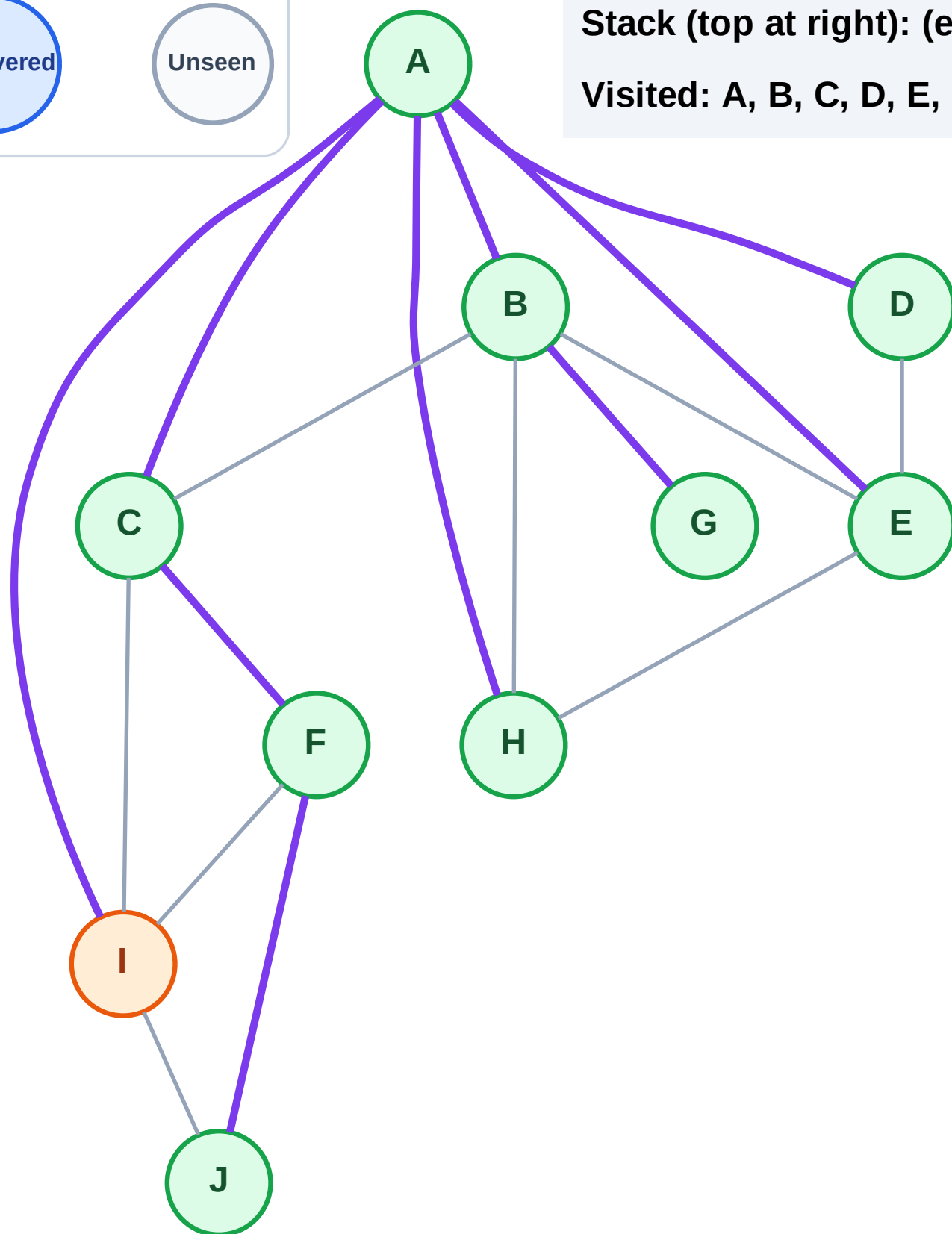
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

